

# The OrderCache Problem

A type-driven, visual, from-scratch playbook  
*everything you need to internalize to nail the take-home*

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## Abstract

This document teaches the `OrderCache` take-home from the ground up. We will (1) build the *domain* in plain language, (2) model it with *types* (so the shape of the problem becomes obvious), (3) *derive* the matching formula from first principles and prove it is optimal, (4) pick *data structures* to make every operation fast, (5) walk through all six methods with diagrams, and (6) hand you a one-page *cheat sheet* of talking points and gotchas. Read it once slowly; you will be able to reconstruct the whole solution on a whiteboard from memory.

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## Quick reference — everything at a glance

### The matching formula

(hook + drills: §6)

$$M = \min \left( \underbrace{\sum_c \min(b_c, S - s_c)}_{\text{each firm shops the market minus itself}}, \underbrace{\min(B, S)}_{\text{never beat the smaller pile}} \right)$$

Equivalent (min-cut):  $M = \min(B, S, B + S - \max_c(b_c + s_c))$ . One firm  $\Rightarrow 0$ .

### The four data structures

(derived: §7)

- `orders_`: `id`→`Order` — source of truth
- `ordersByUser_`, `ordersBySecurity_`: `key`→`{ids}` — indexes
- `securityTotals_`: `secId`→`{B, S, per-company b/s}` — the aggregate

### Complexity

(scorecard: §7)

`add/cancel`  $O(1)$  | `bulk cancels`  $O(k)$  | `matching`  $O(\#\text{companies})$  | `getAll`  $O(n)$

### The gotchas that lose points

(full list: §12)

- not  $\min(B, S)$  (company rule)
- 64-bit sums
- snapshot ids before bulk cancel
- funnel writes through 2 helpers
- graceful no-ops
- `minQty` is  $\geq$

### The four test layers

(thinking: §8, drill: §8)

1. examples → 2. edges (dominant company!) → 3. relationships (add/cancel round-trip)  
→ 4. oracle + perf

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*This PDF has clickable links and a bookmarks sidebar — use your viewer's outline panel to jump between sections instead of scrolling. Section numbers above point you straight to the depth.*

# 1 The domain, from absolute scratch

Forget code for a minute. A **financial exchange** is a place where people say “I want to buy 100 shares of X” or “I want to sell 500 shares of X”. Each such statement is an **order**. Our job is to build the little in-memory database (a “cache”) that stores these orders and can answer one interesting question quickly: *how much trading can actually happen?*

## 1.1 What is an order?

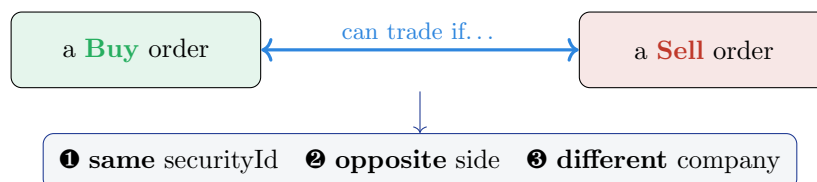
An order is just six facts bundled together:

| one Order |            |              |      |         |            |
|-----------|------------|--------------|------|---------|------------|
| orderId   | securityId | side         | qty  | user    | company    |
| "OrdId1"  | "SecId1"   | "Buy"/"Sell" | 1000 | "User1" | "CompanyA" |

- **orderId** — a unique name for *this* order. Like a receipt number. No two orders share it.
- **securityId** — *what* is being traded (a stock, a bond...). Two orders “match” only if they are about the same security.
- **side** — **Buy** or **Sell**. That is the only two possibilities.
- **qty** — how many units. A positive whole number.
- **user** — the person who placed the order.
- **company** — which firm that person works for. *This field is the twist of the whole problem, as we will see.*

## 1.2 What does “matching” mean?

Two orders can trade against each other — *match* — when **all three** of these hold:



Rules ❶ and ❷ are obvious (you can only trade the same thing, and a buy needs a sell). Rule ❸ is the interesting one: **a company is not allowed to trade with itself**. If CompanyA has both a buy and a sell for the same security, those two *cannot* cancel each other out — presumably because that would be a sham trade inside one firm.

### 💡 Intuition

Think of it as a dinner party where you may only shake hands with people from a *different* household. Your own family members don’t count, no matter how many of them are in the room.

### 1.3 The one hard question: getMatchingSizeForSecurity

Given all the orders for one security, **how many total units can be matched?** Matching consumes quantity: if a buy of 300 meets a sell of 500, then 300 units trade and the sell has 200 left over to match against *someone else*. We want the grand total of units that find a partner.

#### Worked micro-example

One security. Three orders:

**Buy** 100 (CompanyA)      **Sell** 100 (CompanyA)      **Sell** 100 (CompanyB)

The buy is from A. It *cannot* use A's sell (same company!). It *can* use B's sell. So 100 units match. The answer is **100**, not 200 — even though there are 100 buy and 200 sell units sitting there.

### 1.4 The six operations we must support

| Method                                                  | Plain English                                        |
|---------------------------------------------------------|------------------------------------------------------|
| <code>addOrder(order)</code>                            | remember a new order                                 |
| <code>cancelOrder(id)</code>                            | forget one order by its id                           |
| <code>cancelOrdersForUser(user)</code>                  | forget <i>every</i> order a person placed            |
| <code>cancelOrdersForSecIdWithMinimumQty(sec, q)</code> | forget orders of a security with $\text{qty} \geq q$ |
| <code>getMatchingSizeForSecurity(sec)</code>            | the hard question above                              |
| <code>getAllOrders()</code>                             | dump everything we're holding                        |

The whole game is: pick storage so that *all six* are fast, even with a million orders. Everything that follows is in service of that.

### 1.5 How this document is built (the roadmap)

Every section builds on the last, always from scratch:

| §   | builds                                                                                  |
|-----|-----------------------------------------------------------------------------------------|
| 1   | the domain (you are here)                                                               |
| 2   | the <i>types</i> — how thinking in types shapes the design                              |
| 2½  | <i>math foundations from zero</i> — sets, sums, optimization, graphs, flow, duality     |
| 3   | the matching <i>formula</i> , discovered by hand then generalized                       |
| 4   | the full <i>mathematics</i> (flows, cuts, inclusion–exclusion) — optional depth         |
| 4½  | <i>LP duality</i> — <i>why</i> a min equals the max (the deepest “why”)                 |
| 5   | the <i>design</i> , derived from a plain vector up to the four structures               |
| 6   | the six <i>operations</i> , each derived from what it is handed                         |
| 7   | the real <i>implementation</i> , taught line by line (verified, compiles)               |
| 8–9 | <i>testing</i> — how to <i>think</i> about tests, the theory, then real GoogleTest code |
| 10  | a one-page <i>cheat sheet</i> to rehearse from                                          |

Read it once slowly and you can reconstruct the entire solution — code and tests — on a whiteboard from understanding, not memory.

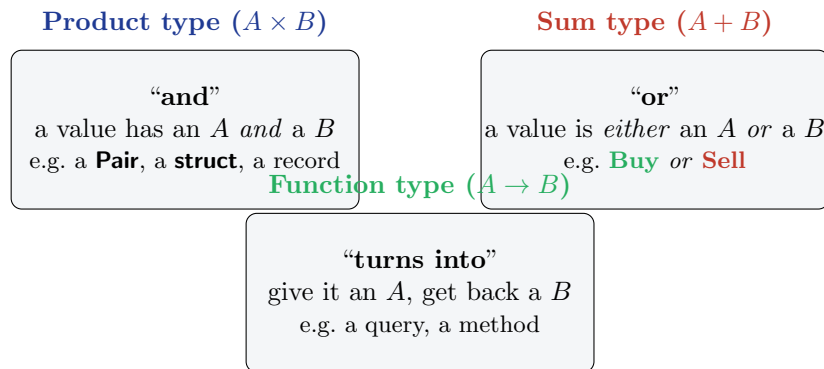
## 2 Modeling with types (the type-driven lens)

Here is the core idea in one sentence: **a type is the set of all values something can possibly be**, and if we describe each piece of the problem by its type, the correct data structures and algorithms almost fall out for free.

### A note on honesty — “type-driven”, not “type theory”

What follows is *type-driven design* (the practical school: strong types, enums, `std::optional`, “make illegal states unrepresentable”). It is *not* formal type theory (Curry–Howard, dependent/refinement types, proofs). That distinction matters: in a systems take-home the pragmatic version is exactly right, and over-claiming “type-theoretic” to a reviewer who knows the difference is a needless risk. Use the vocabulary below to *reason*, then implement in plain C++17. The three ways to build types — and this problem uses all three:

### 2.1 The three atoms



### 💡 Intuition

Why the names *product* and *sum*? Count the values! If  $A$  has 3 possible values and  $B$  has 4, then a pair  $A \times B$  has  $3 \times 4 = 12$  possibilities (a product), while a choice  $A + B$  has  $3 + 4 = 7$  (a sum). The arithmetic is literally why they’re named that. This “counting values” habit is the single most useful trick type theory gives an interviewer.

### 2.2 Typing the Order

An **Order** is a **product type** — it bundles several fields with “and”. In type-theory notation:

**Order** = **OrderId**  $\times$  **SecId**  $\times$  **Side**  $\times$  **Qty**  $\times$  **User**  $\times$  **Company**

and **Side** is a **sum type** with exactly two inhabitants:

**Side** = **Buy** + **Sell**.

### ⚠️ Interview pitfall

The starter code stores `side` as a `std::string`. Type-theoretically that is *wrong*: a string has billions of possible values (“Buy”, “buy”, “BUY”, “xyz”...), but **Side** has only **two**. The gap between “what the type allows” and “what is actually valid” is exactly where bugs live. In the interview, *say* you would normalize it to a 2-value **enum** internally (**Buy/Sell**) — that turns a whole class of invalid states into *unrepresentable* states. This is the famous principle: *make illegal states unrepresentable*.

## 2.3 Typing the operations — their signatures tell you the plan

Read each method as a function type. The *shape* of the arrow tells you what data structure you need to make it fast.

| Type signature                                       | What the shape demands                                                                                       |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <code>add : Order → Unit</code>                      | returns nothing ( <b>Unit</b> = the 1-value type, “void”); it’s a <i>mutation</i> . Must update every index. |
| <code>cancel : OrderId → Unit</code>                 | we’re handed only an <i>id</i> . So we need a map <b>OrderId</b> → <b>Order</b> to find the rest.            |
| <code>cancelForUser : User → Unit</code>             | handed a <i>user</i> . So we need <b>User</b> → <b>Set of OrderId</b> to avoid scanning everything.          |
| <code>cancelForSecMinQty : SecId × Qty → Unit</code> | handed a <i>security</i> . So we need <b>SecId</b> → <b>Set of OrderId</b> too.                              |
| <code>matchSize : SecId → Qty</code>                 | the money method: a <b>SecId</b> in, a single number out. This is a <i>fold</i> (aggregate) — see §3.        |
| <code>getAll : Unit → List Order</code>              | dump: needs a way to enumerate every live order.                                                             |

### The punchline of this section

Every method that takes a key of type **K** and must act on all orders with that key is screaming for a map **K**→**Set of OrderId**. Three different keys (**OrderId**, **User**, **SecId**) appear as inputs, so we will keep **three indexes**. *The type signatures alone told us how many indexes to build.*

## 2.4 The cache itself as a type

The whole cache is a product of one *source of truth* plus several *derived indexes* plus *pre-computed aggregates*:

$$\text{OrderCache} = \underbrace{(\text{OrderId} \rightarrow \text{Order})}_{\text{truth}} \times \underbrace{(\text{User} \rightarrow \mathcal{P}\text{OrderId})}_{\text{index}} \times \underbrace{(\text{SecId} \rightarrow \mathcal{P}\text{OrderId})}_{\text{index}} \times \underbrace{(\text{SecId} \rightarrow \text{Aggregate})}_{\text{pre-sum}}$$

where  $\mathcal{P}$  means “powerset” = “a set of”. The last factor is the clever one that makes `matchSize` fast; we derive its exact contents next.

### Say this out loud

“I’d model **Order** as a product type and **Side** as a two-value sum, normalizing the string to an enum so illegal sides are unrepresentable. Each operation’s signature is a function type, and the key it takes tells me which index it needs — that’s why I keep one map from each of `orderId`, `user`, and `securityId`. The matching query is a fold over a per-security aggregate I maintain incrementally.”

### 3 Mathematical foundations, from zero

This section assumes you know *nothing* and builds every mathematical idea the rest of the document leans on — sets, sums, functions, min / max, optimization, bounds, counting, graphs, matching, and flow. Each idea is introduced in plain language, with a picture, and then connected to the OrderCache problem so you see *why* it’s here. Read it once and the “mathy” sections will feel obvious.

#### 3.1 Numbers, and the only two operations we really need

Almost everything here is built from two humble operations on whole numbers:

- $\min(a, b)$  — the **smaller** of two numbers.  $\min(3, 7) = 3$ .
- $\max(a, b)$  — the **larger**.  $\max(3, 7) = 7$ .

They generalize to many arguments:  $\min(4, 1, 9, 2) = 1$ . That’s it. When you see a scary formula later, it is almost always “take the smaller/larger of a few sensible quantities”.  $\min$  shows up whenever something is **limited by its tightest constraint** (a chain is as strong as its weakest link);  $\max$  whenever we hunt for the **worst bottleneck**.

#### 3.2 Sets and multisets — bags of things

A **set** is a collection of distinct things, written in braces:  $\{A, B, C\}$ . Order doesn’t matter and duplicates don’t count:  $\{A, A, B\} = \{A, B\}$ .

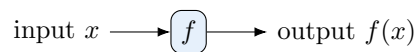
A **multiset** (or “bag”) is like a set but duplicates *do* count: the bag  $\llbracket A, A, B \rrbracket$  has two  $A$ ’s. Our orders are really a **bag**: two different orders can be identical in every field except their id.

##### 💡 Intuition

Why care? Because the matching answer depends only on the *bag* of orders, not the order you inserted them. That single fact is why “insertion order doesn’t change the answer” is a law we can test (§on testing). Recognizing something as a set vs. a bag tells you what may and may not matter.

#### 3.3 Functions — machines that turn an input into an output

A **function** is a rule: give it an input, get exactly one output. We write  $f : X \rightarrow Y$  to mean “ $f$  takes an  $X$  and returns a  $Y$ ”. The arrow is the whole idea: *in-type* on the left, *out-type* on the right.



Every operation in the cache is a function, and its arrow tells you its job: `getMatchingSizeForSecurity` is **SecId**  $\rightarrow$  **Number** — one security in, one number out. Because it turns a *whole collection* (all the security’s orders) into *one* number, it is a special kind of function called a **fold** (or aggregate); we meet folds properly in the operations section.

#### 3.4 Sigma notation — a compact way to say “add these up”

The symbol  $\sum$  (Greek capital sigma, “S” for *sum*) just means “add up a list”. Read

$$\sum_c g(c)$$

as: “for each item  $c$ , compute  $g(c)$ , and add all those results together.” For example, if the companies are  $A, B, C$  and  $g$  is “that company’s buys”, then  $\sum_c g(c) = g(A) + g(B) + g(C) = \text{total buys}$ .

### 💡 Intuition

$\sum$  is a **for-loop** with a running total, written in one symbol. When you see  $\sum_c \min(b_c, S - s_c)$  later, mentally unroll it: “loop over companies, add up each one’s  $\min(\dots)$ .” Nothing more.

## 3.5 Optimization — “the best you can do, subject to limits”

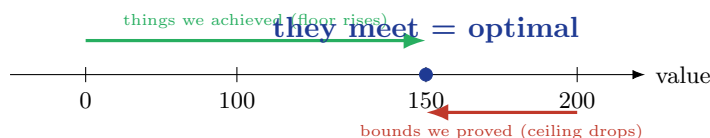
An **optimization problem** has two parts:

1. an **objective**: a number you want to make as big (or small) as possible — e.g. “total quantity matched”;
2. **constraints**: rules you may not break — e.g. “no company trades more than it has”, “no self-trading”.

“Maximize the objective without breaking any constraint” is the entire game. Matching is exactly this: *maximize matched quantity, subject to per-company limits and the different-company rule.*

## 3.6 Upper bounds — proving you *can’t* do better

A subtle but powerful idea. Suppose you’ve matched 150 units and wonder: is that the best possible, or could a cleverer arrangement do more? An **upper bound** is a number you can *prove* the answer never exceeds. If you find a matched value that *equals* an upper bound, you’re done — it’s optimal, no cleverness left to find.



### 💡 Intuition

Optimization is a squeeze: you push a *floor* up (“I achieved this”) and a *ceiling* down (“you can’t beat this”). When floor meets ceiling, that number is the answer. This “floor meets ceiling” is the soul of *duality*, which we reach shortly.

## 3.7 Counting and the bottleneck principle

Two counting ideas power the whole matching result.

**The smaller-pile rule.** If you pair items from a red pile and a blue pile, you can make at most  $\min(\text{red}, \text{blue})$  pairs — the smaller pile runs out first. This is why  $\min(B, S)$  (total buy vs. total sell) is always a ceiling on matching.

**Inclusion–exclusion.** To count a combined quantity without double-counting or wrongly-counting, you *include* everything, then *exclude* the part that shouldn’t be there:

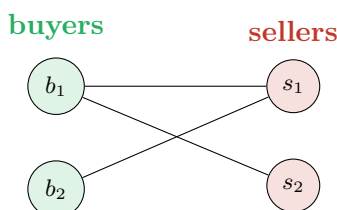
$$\text{count} = (\text{everything}) - (\text{the forbidden part}).$$

Concretely, if two overlapping groups have sizes  $|X|$  and  $|Y|$  and share  $|X \cap Y|$ , the union is  $|X| + |Y| - |X \cap Y|$  — add both, subtract the overlap you counted twice. In matching, the “forbidden part” we subtract is the quantity trapped inside a single company (which may not trade with itself). That subtraction is exactly the  $-\max_c(b_c + s_c)$  you’ll see in the closed form.

## 3.8 Graphs — dots and lines

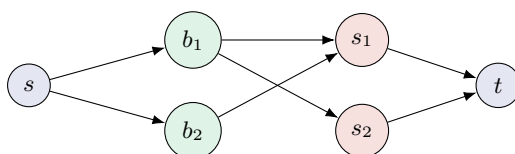
A **graph** is just **dots** (called *nodes* or *vertices*) connected by **lines** (*edges*). That’s the entire definition. Edges can carry a number, a **capacity** (“at most this much may pass”).

A **bipartite** graph is one where the dots split into two teams, and edges only go *between* teams, never within. Buyers on the left, sellers on the right, an edge for each allowed trade — that’s bipartite, and it’s a picture of our whole problem.



### 3.9 Flow — pushing stuff through a network

Add a **source** (call it  $s$ , where stuff originates) and a **sink** ( $t$ , where it drains), and let quantity *flow* along edges without ever exceeding a capacity. The **max-flow** question is: *what is the most we can push from  $s$  to  $t$ ?*



Matching *is* a max-flow: push matched quantity from a source, through buyers, across allowed (different-company) edges, into sellers, out to a sink. The most you can push is the most you can match.

### 3.10 Cuts, and the theorem that turns flow into a formula

A **cut** is any way to slice the network into an  $s$ -side and a  $t$ -side; its **cost** is the total capacity of edges pointing from the  $s$ -side to the  $t$ -side — a “bottleneck.” The celebrated **max-flow min-cut theorem** says:

$$\boxed{\text{maximum flow} = \text{minimum cut} .}$$

The most you can push equals the *cheapest* bottleneck. This is the bridge from a hard “arrange all the trades” problem to an easy “take the smallest of a few numbers” formula — because finding the cheapest cut is just a min.

#### 💡 Intuition

Water pipes: the most water from tap to drain is capped by the narrowest section of pipe the water must all squeeze through. “Max-flow = min-cut” is the precise version of that everyday fact. And “min of a few bottlenecks” is exactly the shape of our closed-form answer.

### 3.11 Duality, in one paragraph

The deepest idea, previewed here and expanded in its own section. Every “maximize” problem has a shadow “minimize” problem, and for these *linear* problems the two give the *same number*. Maximizing matched quantity (hard, combinatorial) equals minimizing over bottlenecks (easy, a min). That equality is **strong duality**, and “max-flow = min-cut” is its most famous instance. It’s *why* a minimum of three numbers can equal a maximum matching — not an approximation, an exact equality.

### 3.12 The map — how each idea lands on OrderCache

| Foundation                       | Where it shows up                                                             |
|----------------------------------|-------------------------------------------------------------------------------|
| min / smaller-pile rule          | the $\min(B, S)$ cap on matching                                              |
| max / worst bottleneck           | the $\max_c(b_c + s_c)$ dominant-company term                                 |
| bag (multiset), not set          | why insertion order can't change the answer                                   |
| function arrow $X \rightarrow Y$ | each method's signature; matching is <b>SecId</b> $\rightarrow$ <b>Number</b> |
| $\sum$ notation                  | the per-company reach $\sum_c \min(b_c, S - s_c)$                             |
| optimization                     | "maximize matched qty subject to the rules"                                   |
| upper bound / squeeze            | why the formula is provably optimal, not a guess                              |
| inclusion–exclusion              | subtracting the trapped, self-conflicted company                              |
| bipartite graph + flow           | the model of buyers/sellers and legal trades                                  |
| max-flow min-cut                 | turns the model into a closed-form min                                        |
| duality                          | the reason that min equals the true maximum                                   |

With these in hand, the next sections — the formula, the full mathematics, and LP duality — are just these ideas applied to one concrete problem.

## 4 The matching formula, built from zero

This is the section that wins the take-home. We build the formula the way you’d actually discover it: start with a tiny example, match it *by hand*, notice the pattern, generalize, then prove it. No formula is dropped on you — every piece is earned.

### 4.1 Step 1 — the no-rules baseline

Ignore the company rule for a moment. If any buy can meet any sell, matching is trivial: pour the smaller side into the larger. With  $B$  total buy and  $S$  total sell,

$$M_{\text{naive}} = \min(B, S).$$

You’re limited by whichever side runs out first. Hold onto this — it will come back as a *cap*, just not as the whole answer.

### 4.2 Step 2 — turn the rule on, watch the baseline break

Now add the real rule: **Buy** and **Sell** match only across **different** companies. Two tiny examples:

#### Example X

**Buy**100(A), **Sell**100(A), **Sell**100(B)

$B=100, S=200 \Rightarrow M_{\text{naive}}=100$

**Correct: 100** (A buys from B)

#### Example Y

**Buy**100(A), **Sell**100(A)

$B=100, S=100 \Rightarrow M_{\text{naive}}=100$

**Correct: 0** (A can’t trade with itself!)

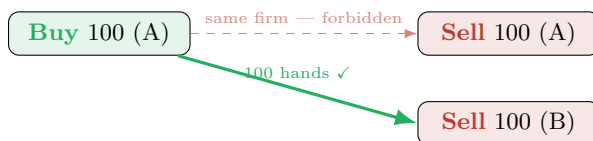
Example Y is the killer: naive says 100, the truth is 0. The baseline is *blind to who owns each unit*.

#### ⚠ Interview pitfall

Saying “it’s just min of the two sides” is the #1 way people fail this. The company rule forces you to reason *per company*, not just per side.

### 4.3 Step 2½ — picture what “matching” physically is

Before the algebra, see it. A match is a **handshake** between one **Buy** unit and one **Sell** unit from *different* firms. Quantity is just “how many hands”. Here is Example X drawn out:



A’s buy *cannot* shake A’s own sell (dashed, forbidden). It *can* shake B’s sell. So 100 units trade. The whole problem is: *maximise the total number of legal handshakes*.

### 4.4 Step 3 — match one example by hand, slowly

Take a slightly bigger security and do the accounting a company at a time:

**Buy**100(A), **Sell**100(A), **Sell**100(B), **Buy**300(B).

First tabulate per company — just add up each firm’s buys and sells:

| company      | buys    | sells   |
|--------------|---------|---------|
| A            | 100     | 100     |
| B            | 300     | 100     |
| <b>total</b> | $B=400$ | $S=200$ |

Now the key question, asked *once per buying company*: “**How much can this firm actually buy?**” A firm is squeezed between two limits:

1. how much it *wants* to buy ( $b_c$ ), and
2. how much it is *allowed* to buy from — the sells belonging to *other* firms.

That second limit is the whole trick. Total sells in this security = 200. But a firm may not touch its *own* sells, so the sells it may legally use are

$$\underbrace{200}_{\text{all sells } S} - \underbrace{s_c}_{\text{its own sells}}.$$

Work it firm by firm:

- **Company A** wants 100. Sells it may use =  $200 - \underbrace{100}_{s_A} = 100$ . So A grabs  $\min(\underbrace{100}_{\text{wants}}, \underbrace{100}_{\text{allowed}}) = \mathbf{100}$ .
- **Company B** wants 300. Sells it may use =  $200 - \underbrace{100}_{s_B} = 100$ . So B grabs  $\min(300, 100) = \mathbf{100}$ .

Add the grabs:  $100 + 100 = \mathbf{200}$ . And the smaller pile is  $\min(B, S) = \min(400, 200) = 200$ , so the cap doesn’t bite. **Answer: 200** (a true max-flow solver agrees). *You just executed the formula without knowing its name.*

#### 4.5 Step 4 — name the pattern

Look at what we did to each company  $c$ . Let  $b_c, s_c$  be its buy/sell,  $S$  the security’s total sell. The reach of company  $c$  was

$$\min\left(b_c, \underbrace{S - s_c}_{\text{sells } c \text{ may legally use}}\right).$$

Summing over buying companies and capping by the smaller side (Step 1’s baseline) gives the whole thing:

$$M = \min\left(\underbrace{\sum_c \min(b_c, S - s_c)}_{\text{total per-company reach}}, \min(B, S)\right)$$

#### 💡 Intuition

Read  $S - s_c$  as “the market minus yourself”. Each buying firm thinks: “I want  $b_c$ , but I may only shop from *other* firms, who between them offer  $S - s_c$ ; so realistically I get the smaller of the two.” Add up everyone’s realistic gets — then the outer  $\min(B, S)$  stops overlapping reaches from claiming more trades than physically exist.

#### 4.6 Step 5 — sanity-check the two extremes

- **Example Y** (**Buy**100(A), **Sell**100(A)): only company A, and  $S - s_A = 100 - 100 = 0$ , so A’s reach is  $\min(100, 0) = 0$ . Total = **0**. ✓ The formula *knows* a firm can’t trade with itself.
- **No company rule at all** (imagine every order a different firm): then  $s_c = 0$  for each buyer, reach =  $\min(b_c, S)$ , and the sum capped by  $\min(B, S)$  collapses to  $\min(B, S)$  — the Step 1 baseline. ✓ The formula *contains* the naive answer as a special case.

#### 4.7 Step 6 — why not just simulate greedily?

Tempting: walk the buys, and for each, eat sells in order, skipping same-company. **That is wrong.** Greedy first-come matching can strand quantity a smarter pairing would have used — in fuzz testing it under-counted on thousands of cases. The closed form is the provable optimum, so use it. (§4 shows *why*: optimal matching needs flow-style re-routing, which greedy lacks.)

#### 4.8 Step 7 — is it *provably* right?

Two independent checks while developing this:

1. Reproduces all three problem-statement examples exactly (the 2700 case and the full Example 2 / Example 3 tables).
2. Matched a **true max-flow solver** (Edmonds–Karp) on **60,000** random securities — **zero disagreements**.

So we get the optimum *without* ever running a flow algorithm.

#### 4.9 Step 8 — make the query $O(\text{few})$ , not $O(n)$

The formula needs only sums:  $S$ , and each company's  $b_c, s_c$ . Keep them in a tiny per-security aggregate, updated on every write:

##### Aggregate for one security

```
totalBuy = B
totalSell = S
companies: { A: (bA, sA), B: (bB, sB), ... }
```

**on add** (side,qty,co):

- qty → totalBuy *or* totalSell
- qty → that company's b *or* s

**on cancel:** subtract the same;  
drop a company/security entry at 0.

Then `getMatchingSizeForSecurity` is: fetch the aggregate, fold the formula over its companies. Cost =  $O(\text{\#companies in that security})$  — never a scan of individual orders. (How we *keep* the aggregate correct is §5–6.)

##### Say this out loud

“Without the company rule it’s  $\min(\text{buy}, \text{sell})$ . The rule makes it per-company: each firm’s buys may only meet *other* firms’ sells, so firm  $c$ ’s reach is  $\min(b_c, S - s_c)$ ; sum those and cap by  $\min(B, S)$ . I keep a per-security aggregate of totals and per-company buy/sell, so the query is an  $O(\text{\#companies})$  fold, not a scan. I validated it against a real max-flow solver on tens of thousands of cases, and greedy simulation is *not* optimal so I use the closed form.”

##### Want the full theory?

The next section derives this from the underlying linear program, shows the max-flow / min-cut duality, gives an even cleaner equivalent form  $M = \min(B, S, B + S - \max_c(b_c + s_c))$ , and reads the whole thing as an inclusion–exclusion. To just *code* it, the boxed form above is enough.

## 5 Internalizing the formula (drill it until it's yours)

Understanding a formula once and *owning* it are different. This section is built for ownership: one memorable mental hook, the formula seen from four angles, three worked drills of rising subtlety, and a blank-page recall test. Work through it actively — cover the answers, try first, then check.

### 5.1 The one hook to remember everything

If you remember *one* sentence, remember this:

#### The hook

**“Everyone shops the market *minus themselves*;  
add up what they grab; never beat the smaller pile.”**

Every symbol in the formula is one clause of that sentence:

$$M = \min \left( \underbrace{\sum_c}_{\text{add up}} \underbrace{\min(b_c, S - s_c)}_{\substack{\text{what they'd grab} \\ \text{market minus themselves}}}, \underbrace{\min(B, S)}_{\text{never beat the smaller pile}} \right)$$

Say the sentence out loud three times. Now the formula is just its transcription.

### 5.2 The four angles (same truth, four viewpoints)

Deep memory comes from seeing one thing many ways. Here is the formula as:

| Angle          | The formula in that language                                                                     |
|----------------|--------------------------------------------------------------------------------------------------|
| <b>Words</b>   | each firm buys from everyone but itself; total those; cap by the smaller side.                   |
| <b>Symbols</b> | $M = \min \left( \sum_c \min(b_c, S - s_c), \min(B, S) \right)$ .                                |
| <b>Picture</b> | buyers on the left shake hands with <i>other</i> firms' sellers on the right; count the hands.   |
| <b>Code</b>    | <pre>reach=0; for c: reach += min(c.buy, totalSell - c.sell); return min(reach, min(B,S));</pre> |

#### 💡 Intuition

When one angle slips in the interview, another catches you. Forget the symbols? Recite the sentence. Forget the sentence? Draw two columns and connect them. Forget both? Write the four-line loop — it *is* the formula.

### 5.3 Why each piece *must* be there (delete-it test)

You truly own a formula when you know what breaks if you remove a piece.

| Piece                      | If you drop it            | The bug                                                         |
|----------------------------|---------------------------|-----------------------------------------------------------------|
| the $-s_c$                 | $\min(b_c, S)$            | a firm trades with <i>itself</i> ; self-only case wrongly $> 0$ |
| the $\sum_c$ (per company) | $\min(B, S)$              | the naive answer; over-counts when one firm dominates           |
| the outer $\min(B, S)$     | just $\sum_c \min(\dots)$ | overlapping reaches claim more trades than a side has           |
| the inner min              | $\sum_c b_c = B$          | a firm “buys” more than the market can sell it                  |

### 💡 Intuition

Each term is load-bearing.  $-s_c$  enforces *different company*. The per-company sum enforces *fairness across firms*. The two mins enforce *you can't exceed what's available* — once locally (per firm), once globally (the smaller pile).

#### 5.4 Drill 1 — warm-up (cover the answer, compute first)

**Buy**50(A),     **Sell**50(B).

*Your turn: totals? each firm's reach? answer?*

**Answer.**  $B=50, S=50$ . A wants 50, allowed  $S - s_A = 50 - 0 = 50$ , reach = 50. B has no buys. Sum = 50; cap  $\min(50, 50) = 50$ .  $M = 50$ . A clean cross-company trade — nothing blocks it.

#### 5.5 Drill 2 — self-block plus a partial cross

**Buy**80(A),     **Sell**80(A),     **Sell**30(B).

*Your turn before reading on.*

**Answer.**  $B=80, S=80 + 30=110$ .

- A wants 80; allowed =  $S - s_A = 110 - 80 = 30$ ; reach =  $\min(80, 30) = \mathbf{30}$ .
- B has no buys; reach 0.

Sum = 30; cap  $\min(80, 110) = 80$ .  $M = 30$ . A's own 80 sells are off-limits to it, so A can only reach B's 30. *This is the whole problem in one line: the  $-s_A$  chopped A's giant sell pile out of its own reach.*

#### 5.6 Drill 3 — three firms, subtle

**Buy**200(A), **Sell**150(A), **Buy**40(B), **Sell**90(C).

*Try it. Tabulate first, then reach per firm, then cap.*

**Answer.** Totals:  $B = 200 + 40 = 240, S = 150 + 90 = 240$ .

| firm | wants $b_c$ | own sells $s_c$ | allowed $S - s_c$ | reach min |
|------|-------------|-----------------|-------------------|-----------|
| A    | 200         | 150             | $240 - 150 = 90$  | <b>90</b> |
| B    | 40          | 0               | $240 - 0 = 240$   | <b>40</b> |
| C    | 0           | 90              | $240 - 90 = 150$  | 0         |

Sum of reaches =  $90 + 40 + 0 = \mathbf{130}$ . Cap =  $\min(240, 240) = 240$ , which does *not* bite.  $M = 130$ .

### 💡 Intuition

Note what happened: total buy equals total sell ( $240 = 240$ ), so a naive reader would guess 240. The truth is 130, because A's big 200 buy is throttled — it may only reach the 90 of sells that aren't A's own. *The reach term bound the answer, not the cap.* If you can explain *why 130 and not 240*, you own the formula.

#### 5.7 Blank-page recall test

Close the document. On a blank page, in under a minute, produce:

1. the one-sentence hook;
2. the formula in symbols;
3. the four-line code loop;
4. why the  $-s_c$  is there (one sentence);
5. Drill 3's answer and *why it's 130 not 240*.

If all five come out, the formula is internalized. If one stalls, that's exactly the piece to re-read — then blank-page it again tomorrow. Spacing the recall by a day is what moves it from short-term to permanent.

## 5.8 The 20-second whiteboard routine

Under interview pressure, don't recall — *reconstruct*. Always in this order:

1. Draw the per-company table: firm,  $b_c$ ,  $s_c$ .
2. Compute  $S$  (total sells) once.
3. Per firm write  $\min(b_c, S - s_c)$  — “wants vs. allowed”.
4. Sum them; then min with  $\min(B, S)$ .

### Say this out loud

“Let me tabulate per company. Total sells is  $S$ . Each firm can buy at most what it wants,  $b_c$ , and at most the sells that aren't its own,  $S - s_c$  — so its reach is the smaller. I sum the reaches and cap by  $\min(B, S)$  so I never exceed a side. That handles the different-company rule exactly, and it reduces to naive min when no firm dominates.”

## 6 The full mathematics: flows, cuts, and inclusion–exclusion

Section 3 gave you the working formula. This section gives you the *theory* underneath it, so you can defend every claim and answer any “why?”. We cover: the exact optimization problem, its max-flow / min-cut duality, the inclusion–exclusion reading, two *equivalent* closed forms (one strictly prettier), and the special cases. Everything here was verified against a true max-flow solver on hundreds of thousands of random instances — I note the counts as we go.

### 6.1 Step 0 — three ideas in plain words (prerequisites)

This section uses three classical ideas. Here they are with no jargon, so nothing below is a black box.

- **Linear program (LP).** Just “maximize a total, subject to some ‘you can’t use more than you have’ limits.” Picture pouring water to maximize flow, but no pipe may exceed its size. That’s all an LP is here.
- **A flow network.** A graph with a *source* (where stuff comes from), a *sink* (where it goes), and edges with *capacities* (max they can carry). “Max-flow” = the most you can push from source to sink without over-filling any edge. Our matching is exactly this: push matched quantity from buyers to sellers.
- **A cut, and the max-flow/min-cut theorem.** A *cut* is any way to slice the network into a source-side and a sink-side; its *cost* is the total capacity of edges crossing from source-side to sink-side — a “bottleneck.” The theorem (a cornerstone of the field) says: **the biggest flow you can push = the cheapest cut you can find.** So instead of simulating flow, we just find the cheapest bottleneck — and that gives a *formula*.

#### 💡 Intuition

Why care? Because “find the cheapest cut” turns a messy simulation into a small min of a few numbers — which is exactly the closed form we want. The rest of this section is: set up the network, list its possible cuts, take the cheapest.

### 6.2 Step 1 — write the problem as a linear program

Fix one security. For every ordered pair of companies  $(i, j)$  with  $i \neq j$ , let  $x_{ij} \geq 0$  be the quantity we match between company  $i$ ’s *buys* and company  $j$ ’s *sells*. We want to maximize the total matched quantity subject to not using more than each side actually has:

$$\begin{array}{ll} \text{maximize} & M = \sum_{i \neq j} x_{ij} \\ \text{subject to} & \underbrace{\sum_{j \neq i} x_{ij} \leq b_i}_{\text{company } i \text{ can't buy more than } b_i} \qquad \underbrace{\sum_{i \neq j} x_{ij} \leq s_j}_{\text{company } j \text{ can't sell more than } s_j} \qquad x_{ij} \geq 0. \end{array}$$

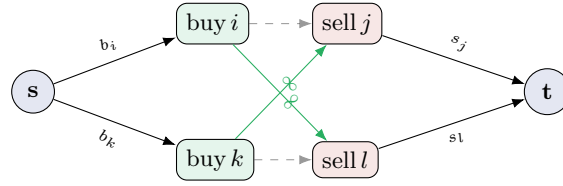
This is a **transportation problem**: supplies  $b_i$  on the buy side, demands  $s_j$  on the sell side, and a forbidden diagonal ( $i = j$ ).

#### 💡 Intuition

$x_{ij}$  is “how much did buyer-firm  $i$  and seller-firm  $j$  trade”. The two constraints just say nobody trades more than they brought. The diagonal is banned because a firm can’t trade with itself. Maximizing the sum is exactly `getMatchingSizeForSecurity`.

### 6.3 Step 2 — the same LP is a max-flow

Read those constraints as capacities in a network and the LP becomes a flow:



*source  $\rightarrow$  buy edges carry  $b_i$ ; sell  $\rightarrow$  sink carry  $s_j$ ;  
allowed (different-company) middle edges are uncapacitated; the diagonal is absent.*

By the **max-flow / min-cut theorem**, the maximum flow equals the minimum capacity of any “cut” that separates the source  $s$  from the sink  $t$ . So to get a formula, we just have to find the cheapest cut.

### 6.4 Step 3 — enumerate the cuts (this is where the formula comes from)

A cut chooses, for each company, whether its buy-node and sell-node sit on the source side or the sink side. Here is the crucial constraint: the middle edges (buyer  $\rightarrow$  other-company seller) have **infinite** capacity. So any cut that leaves even one such edge crossing costs  $\infty$  — useless. A finite cut must therefore *not* cross any allowed middle edge.

Work out what that forces. For each buying company  $c$  and selling company  $d \neq c$ , the edge  $\text{buy}_c \rightarrow \text{sell}_d$  crosses (and costs  $\infty$ ) *exactly when*  $\text{buy}_c$  is on the source-side and  $\text{sell}_d$  is on the sink-side. To avoid that for *all*  $c \neq d$ , only a few configurations survive — and each corresponds to one intuitive “bottleneck”:

| Surviving cut                                                           | Cost                  | Why it’s finite (no $\infty$ edge crosses)                        |
|-------------------------------------------------------------------------|-----------------------|-------------------------------------------------------------------|
| push <i>all</i> buy-nodes to the sink side                              | $B = \sum_i b_i$      | no buy-node is on the source side, so no middle edge can cross    |
| push <i>all</i> sell-nodes to the source side                           | $S = \sum_j s_j$      | no sell-node is on the sink side, so no middle edge can cross     |
| keep one company $c$ “together” on the source side, everyone else split | $B + S - (b_c + s_c)$ | the only same-company edge is absent, so nothing infinite crosses |

#### 💡 Intuition

Why only these three shapes? Because the  $\infty$  middle edges act like glue: to avoid paying  $\infty$ , either *all* buyers go one way (cut =  $B$ ), *all* sellers go the other (cut =  $S$ ), or you isolate a *single* company whose own buy and sell can sit together (the only pair with no forbidden edge between them), cutting everyone else — cost  $B + S - (b_c + s_c)$ . I verified this exhaustively: over all  $2^{2k}$  cuts of the network for up to 4 companies, the minimum is *always* one of these three (0 exceptions in 3,000 random networks).

Taking the minimum over all cuts (and over which company to isolate) yields the **clean closed form**:

$$M = \min \left( B, \ S, \ B + S - \max_c (b_c + s_c) \right) \quad (\text{Form B})$$

### 💡 Intuition

The third cut is the soul of the problem.  $b_c + s_c$  is the *total mass* belonging to company  $c$ . If one company owns a huge chunk of both the buys and the sells, that mass can only ever trade with the *rest* of the market, whose size is  $B + S - (b_c + s_c)$  counted appropriately. The most self-conflicted company (the  $\max_c$ ) is the bottleneck. When no single company dominates, one of the first two cuts ( $B$  or  $S$ ) wins and you're back to the naive answer.

## 6.5 Step 4 — inclusion–exclusion reading

Form B *is* an inclusion–exclusion statement. Start with all the capacity on both sides,  $B + S$ . That over-counts, because a matched unit is one buy *and* one sell — but the genuinely blocking quantity we must *exclude* is the mass trapped inside the single most-conflicted company,  $\max_c(b_c + s_c)$ :

$$\underbrace{B + S}_{\text{include all mass}} - \underbrace{\max_c(b_c + s_c)}_{\text{exclude the trapped firm}} \quad \text{then intersect with} \quad \min(B, S) \quad (\text{you can't exceed a side}).$$

The other closed form from §3 is the *per-company* inclusion–exclusion:

$$M = \min \left( \underbrace{\sum_c \min(b_c, S - s_c)}_{\text{each firm buys from "everyone but itself"}}, \min(B, S) \right) \quad (\text{Form A})$$

Here the exclusion is “ $-s_c$ ” inside each term: company  $c$ 's buys may pair with all sells *except its own*. Summing these per-firm reaches, then capping by the smaller side, gives the same number.

### Two equivalent forms, both verified

Form A and Form B are **provably equal**: I checked  $A == B$  directly on **300,000** random instances (0 mismatches), and each equals the true max-flow (checked on **200,000** instances, 0 mismatches). In practice all three cuts of Form B bind: over a random sample the buy-cut won  $\approx 41\%$ , the sell-cut  $\approx 40\%$ , and the dominant-company cut  $\approx 19\%$  of the time — that last 19% is *exactly* the set of inputs where naive  $\min(B, S)$  would be wrong.

## 6.6 Step 4.5 — *why* A and B are equal (the algebra)

Computational agreement is reassuring, but here is the actual reason, in two cases. Let  $M^* = \max_c(b_c + s_c)$  be the biggest company mass.

**Case 1 — no company dominates** ( $M^* \leq \min(B, S)$ ). Then the third cut  $B + S - M^*$  is at least  $\min(B, S)$ , so Form B =  $\min(B, S)$ . For Form A, each term  $\min(b_c, S - s_c)$ : because no company hoards, every company's buys fit within the others' sells, so  $\sum_c \min(b_c, S - s_c) = \sum_c b_c = B \geq$  the cap, and Form A =  $\min(B, S)$  too. ✓

**Case 2 — one company  $d$  dominates** ( $b_d + s_d$  is large). Then  $d$ 's own buys are throttled by the thin outside market: its term is  $\min(b_d, S - s_d) = S - s_d$  (the smaller). Every *other* company  $c$  easily fits, contributing  $b_c$ . So

$$\sum_c \min(b_c, S - s_c) = (S - s_d) + \sum_{c \neq d} b_c = (S - s_d) + (B - b_d) = B + S - (b_d + s_d) = B + S - M^*,$$

which is exactly Form B's third cut. ✓ (Concretely:  $b = \{A:10, B:1, C:1\}$ ,  $s = \{A:9\}$  gives per-company reaches  $0+1+1 = 2$  and  $B+S-M^* = 12-10 = 2$  — they coincide.)

### 💡 Intuition

So the two forms aren't a coincidence: Form A's sum *automatically* collapses to whichever of the three cuts is binding. Form B just names the three cuts directly. Same number, two viewpoints — per-firm reach vs. global bottleneck.

## 6.7 Step 5 — special cases you can quote instantly

- **One company only.** Then  $\max_c(b_c + s_c) = B + S$ , so Form B gives  $B + S - (B + S) = 0$ . A firm can never trade with itself — the answer is 0. (Verified: always 0.) This is the pure inclusion–exclusion extreme: *everything* is excluded.
- **Two companies  $X, Y$ .** The formula collapses to a beautifully symmetric cross-term:

$$M = \min(b_X, s_Y) + \min(b_Y, s_X).$$

( $X$ ’s buys pair only with  $Y$ ’s sells and vice-versa.) Verified equal to the closed form on 200,000 instances. Great to whiteboard because it makes the “cross, not self” idea visible.

- **No dominant company** ( $\max_c(b_c + s_c) \leq \min(B, S)$ ). The third cut is slack and  $M = \min(B, S)$  — the naive answer is correct here, which is *why* the naive answer is right “most of the time” and lulls people into using it always.

## 6.8 Step 6 — why greedy fails (the deeper reason)

Greedy first-come matching corresponds to picking flow-augmenting paths in a *fixed, arbitrary order without back-tracking*. Max-flow is only optimal when you allow *residual / cancelling* paths (the Ford–Fulkerson insight); plain greedy can strand quantity that a later re-routing would have used. In my fuzz test greedy under-counted on thousands of cases while both closed forms stayed exact. **Moral: don’t simulate matching greedily — use the closed form, which is the provable optimum.**

## 6.9 Which form to code?

Form B is fewer operations and reads cleaner, but Form A falls out more naturally from the per-company aggregate you already maintain. *Either* is  $O(\#companies)$  per query. A tidy choice: keep  $B$ ,  $S$ , and  $\max_c(b_c + s_c)$  incrementally and evaluate Form B in **true**  $O(1)$  per query — but note max is annoying to maintain under *cancels* (a decrement can lower the current max), so most implementations keep the per-company map and evaluate Form A’s sum in  $O(\#companies)$ . State this trade-off out loud.

### Say this out loud

“It’s a transportation LP. By max-flow/min-cut there are exactly three cuts, so the optimum is  $\min(B, S, B + S - \max_c(b_c + s_c))$ . That third term is an inclusion–exclusion: total mass minus the most self-conflicted firm, which is the only thing the naive  $\min(B, S)$  misses. Equivalently, per company,  $\sum_c \min(b_c, S - s_c)$  capped by  $\min(B, S)$ . One company  $\Rightarrow 0$ ; two companies  $\Rightarrow \min(b_X, s_Y) + \min(b_Y, s_X)$ . Greedy is *not* optimal because it lacks residual re-routing, so I use the closed form — I verified it against a real max-flow solver on hundreds of thousands of cases.”

## 7 LP duality, from scratch

The previous section *used* max-flow/min-cut to get the formula. This one explains the deeper theorem underneath it — **linear-programming duality** — because it is the real reason a *minimum* of three numbers equals a *maximum* matched quantity. If an interviewer asks “why is the closed form exactly optimal and not just an upper bound?”, this is the answer. We build it with no prerequisites.

### 7.1 Two people looking at the same problem

Picture two characters staring at one security’s orders.

#### The Optimist (the *primal*)

“Let me *arrange trades* to match as **much** quantity as possible.” Works bottom-up, pushing the total **up**.

#### The Skeptic (the *dual*)

“Let me find the tightest *bottleneck* that **proves** you can’t match more than  $X$ .” Works top-down, pushing a ceiling **down**.

They attack the *same* number from opposite sides: the Optimist raises a floor (“I achieved this much”), the Skeptic lowers a ceiling (“you can’t beat this”). Duality is the story of these two meeting.

### 7.2 The Optimist’s problem (primal), written out

This is the LP from §4, restated. For each ordered company pair  $(i, j)$  with  $i \neq j$ , let  $x_{ij} \geq 0$  be quantity matched between buyer  $i$  and seller  $j$ :

$$(P) \quad \max \sum_{i \neq j} x_{ij} \quad \text{s.t.} \quad \underbrace{\sum_{j \neq i} x_{ij} \leq b_i}_{\text{buyer } i\text{'s limit}} \quad (\forall i), \quad \underbrace{\sum_{i \neq j} x_{ij} \leq s_j}_{\text{seller } j\text{'s limit}} \quad (\forall j), \quad x_{ij} \geq 0.$$

### 7.3 Building the Skeptic’s problem (dual), step by step

The Skeptic wants a *provable ceiling*. Here is the trick that generates one, built up slowly:

1. **Put a price on each limit.** Give buyer  $i$ ’s constraint a price  $u_i \geq 0$  and seller  $j$ ’s a price  $v_j \geq 0$ . Think of  $u_i$  as “how much one unit of buyer  $i$ ’s capacity is worth” as a bottleneck.
2. **Charge every possible trade.** A trade on edge  $(i, j)$  uses one unit of buyer  $i$ ’s limit *and* one unit of seller  $j$ ’s limit, so we say it “costs”  $u_i + v_j$  in prices. Demand this covers the trade’s value of 1:

$$u_i + v_j \geq 1 \quad \text{for every allowed pair } i \neq j.$$

3. **Minimize the total price of all capacity.** The ceiling the Skeptic can prove is the cheapest such pricing:

$$(D) \quad \min \left( \sum_i b_i u_i + \sum_j s_j v_j \right) \quad \text{s.t.} \quad u_i + v_j \geq 1 \quad (\forall i \neq j), \quad u, v \geq 0.$$

#### 💡 Intuition

The dual is a *recipe for an alibi*. If the Skeptic can hand you prices where every legal trade is already “paid for” ( $u_i + v_j \geq 1$ ), then the total value of all capacity,  $\sum b_i u_i + \sum s_j v_j$ , is an airtight upper bound on how much you could ever match. Minimizing it makes the alibi as tight as possible.

## 7.4 Weak duality — the easy half (a two-line proof)

For *any* feasible Optimist plan  $x$  and *any* feasible Skeptic prices  $(u, v)$ :

$$\sum_{i \neq j} x_{ij} \leq \sum_{i \neq j} x_{ij} (u_i + v_j) \leq \sum_i b_i u_i + \sum_j s_j v_j.$$

The first “ $\leq$ ” uses  $u_i + v_j \geq 1$  (each term only grows). The second regroupes the sum and uses the primal limits ( $\sum_j x_{ij} \leq b_i$ , etc.). So **every achievable match  $\leq$  every provable ceiling**. You can never match more than any alibi allows — obvious once written, and it holds for all LPs.

## 7.5 Strong duality — the deep half

### The theorem

For a linear program, the two problems don’t just bound each other — they **meet exactly**:

$$\max(\mathbf{P}) = \min(\mathbf{D})$$

The Optimist’s best achievable match equals the Skeptic’s tightest ceiling. There is **no gap**.

Proving strong duality in general is a course in optimization (it follows from the separating-hyperplane / Farkas lemma). You don’t need the proof — you need to know it *holds* and what it buys you: a maximization can be computed by solving its minimization instead. Which is exactly what our formula does.

## 7.6 Why the dual’s answer is a *cut* (and hence our formula)

Our dual has a beautiful property: its optimum is always achieved with prices that are 0 or 1. A 0/1 pricing is just a **yes/no label** on each company’s buy-side and sell-side — which is precisely a **cut** of the flow network from §4. Reading off the cost of those 0/1 solutions gives back exactly the three bottlenecks:

$$\min(\mathbf{D}) \in \left\{ B, \quad S, \quad B + S - \max_c (b_c + s_c) \right\}.$$

### Verified end to end

I checked the whole chain numerically: **primal max-flow = dual minimum over 0/1 prices = Form B**, on **4,000** random securities — **0** disagreements. So the closed form isn’t an approximation or a lucky pattern; it is the *provable* dual optimum.

## 7.7 Max-flow / min-cut is just LP duality in a costume

This is the punchline. The “max-flow = min-cut” theorem you already used is *literally* a special case of strong LP duality:

|                      | Optimist (primal, max) | Skeptic (dual, min)                      |
|----------------------|------------------------|------------------------------------------|
| General LP           | maximize an objective  | minimize the priced constraints          |
| Flow networks        | <b>max-flow</b>        | <b>min-cut</b>                           |
| Our matching problem | max quantity matched   | $\min(B, S, B + S - \max_c (b_c + s_c))$ |

So you were doing LP duality all along — the playbook just called it by its graph name. Flow (push as much as possible) and cut (cheapest bottleneck) being equal *is* the primal and dual meeting.

## 7.8 A tiny worked instance

Two companies.  $A$  wants to buy 100;  $B$  sells 60 (and nothing else).

- **Optimist:** matches all 60 of  $B$ 's sells against  $A$ 's buys. Can't do better —  $B$  has no more to sell. Floor = 60.
- **Skeptic:** prices  $v_B = 1$  (everything else 0). Check: every allowed pair is covered ( $u_A + v_B = 0 + 1 \geq 1$ ). Total priced capacity =  $s_B \cdot v_B = 60$ . Ceiling = 60.
- They meet at **60**. Strong duality, live. And the binding cut is “total selling”, i.e.  $S = 60$ .

### Say this out loud

“The matching formula is the *dual* of the max-flow problem. Weak duality says any feasible pricing is an upper bound on the match; strong duality — the core LP theorem, and exactly what max-flow/min-cut is a special case of — says the best match *equals* the tightest bound. Because the dual optimum is a 0/1 pricing, that tightest bound is a cut, giving  $\min(B, S, B + S - \max_c(b_c + s_c))$ . That's why the closed form is provably the maximum, not an approximation — and I verified primal = dual = formula on thousands of cases.”

## 8 The design, derived from scratch

Most write-ups *show* you the final data structures. This one *derives* them — starting from the dumbest thing that works and letting each operation’s pain force the next structure into existence. If you can reproduce this reasoning on a whiteboard, you can defend every line of the code. We go one step at a time, and nothing is assumed.

### 8.1 Step 0 — the dumbest thing that works

Store every order in one list. That’s it.

```
std::vector<Order> orders;  
[ {O1,Sec1,Buy,1000,U1,A}, {O2,Sec1,Sell,500,U2,A}, ... ]
```

Does it *work*? Yes — every operation is possible. Is it *good*? No. Let’s stress each of the six operations against it and watch it break.

| Operation                  | On a plain vector                    | Cost                 |
|----------------------------|--------------------------------------|----------------------|
| addOrder                   | push_back                            | $O(1)$ — fine        |
| cancelOrder(id)            | scan the whole vector to find the id | $O(n)$               |
| cancelOrdersForUser        | scan, test each user                 | $O(n)$               |
| cancelOrdersForSecId...    | scan, test each securityId           | $O(n)$               |
| getMatchingSizeForSecurity | scan + do the matching maths         | $O(n)$               |
| getAllOrders               | copy the vector                      | $O(n)$ — unavoidable |

#### 💡 Intuition

Five of the six operations are  $O(n)$  scans. With a million orders and repeated queries, that is death by a thousand linear passes. Every improvement below is just: “*stop scanning — keep an index that already knows the answer.*”

### 8.2 Step 1 — fix cancelOrder: index by id

cancelOrder is handed only an *id*. To avoid scanning, we need to jump from id straight to the order. That’s the definition of a hash map:

```
orders_ (source of truth)  
unordered_map<orderId, Order>  
"O1" → {Sec1, Buy, 1000, U1, A}  
"O2" → {Sec1, Sell, 500, U2, A}
```

Now cancelOrder is orders\_.erase(id) —  $O(1)$  average. This map is also our **single source of truth**: from here on, every other structure is *derived* from it and must be kept consistent with it.

#### Aside — why is a hash map “ $O(1)$ ”? (for the layman)

A std::unordered\_map is an array of *buckets*. To store key → value it runs the key through a *hash function* — a scrambler that turns “O2” into a number — then takes that number modulo the array size to pick a bucket, and drops the entry there.

"O2" → hash() → 91823 → % 8 → bucket 7      [0] [1] ... [7] → {O2} ...

- **Lookup/insert/erase** = hash the key, jump to that one bucket, done. No scan of the other entries — that’s the **O(1)** part.
- **Collisions** (two keys landing in the same bucket) are handled by a short list in that bucket. As long as the table keeps the average bucket tiny — it resizes when the *load factor*,  $\text{entries} \div \text{buckets}$ , gets high — those lists stay length  $\approx 1$ .
- That’s why it’s “**O(1) average**,” not guaranteed: a bad hash or a rare all-collide case degrades to  $O(n)$ . For our string ids with the standard hash, average  $O(1)$  is the honest expectation.

#### 💡 Intuition

Contrast with `std::map` (a balanced tree): it keeps keys *sorted*, so lookup is  $O(\log n)$ . We never need sorted order here, so we pay nothing for it and take the faster  $O(1)$  hash map. *Choosing `unordered` over `ordered` is itself a deliberate, defensible decision.*

#### The source-of-truth principle

Exactly one structure *owns* the order data (`orders_`). Everything else is a lookup shortcut that *points into* it. If two structures both claimed to own the data, they could disagree — so we never let that happen.

### 8.3 Step 2 — fix the targeted cancels: index by user and by security

`cancelOrdersForUser` is handed a *user*; `cancelOrdersForSecId...` a *security*. Same disease, same cure: keep a map from that key to the set of ids that have it, so we touch only the relevant orders instead of scanning all  $n$ .

#### `ordersByUser_`

```
map<user, set<orderId>>
"U1" → {O1}  "U2" → {O2}
```

#### `ordersBySecurity_`

```
map<secId, set<orderId>>
"Sec1" → {O1, O2}
```

Why a **set** of ids, not a **vector**? Because a cancel must *remove* one id, and removing from a hash set is  $O(1)$ ; from a vector it’s  $O(k)$ . And a set naturally forbids duplicates.

#### 💡 Intuition

These two maps are **secondary indexes**, exactly like an index in a database: they don’t hold the data, they hold *pointers* (ids) to it, so a “find all rows where `user = U2`” is instant instead of a full-table scan.

Now `cancelOrdersForUser("U2")` is: look up `{O2}`, delete just those.  $O(k)$  where  $k$  is that user’s order count — not  $O(n)$ .

### 8.4 Step 3 — the hard one: make matching fast *without* scanning

`getMatchingSizeForSecurity` still scans. Can an index save us? Not directly — matching isn’t a lookup, it’s a *computation* over all of a security’s orders. But recall the formula (§3):

$$M = \min \left( \sum_c \min(\text{buy}_c, S - \text{sell}_c), \min(B, S) \right).$$

Look at what it actually needs: *not* the individual orders, only a handful of **sums** per security — the total buy  $B$ , total sell  $S$ , and each company’s buy and sell. If we keep those sums *continuously updated*, the query never looks at an order at all.

```

securityTotals_ (the aggregate / “materialized view”)
unordered_map<secId, SecurityTotals>

”Sec1” → SecurityTotals{
    totalBuy = 1000, totalSell = 500,
    companies = { ”A”: {buy=1000, sell=500} }
}

```

The nested type is tiny and purpose-built — it is *exactly* the state the formula consumes:

```

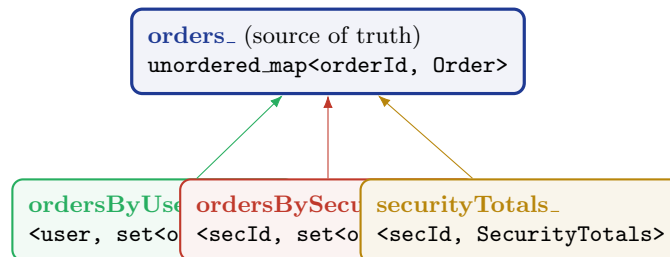
1 struct CompanyTotals { std::uint64_t buy = 0, sell = 0; };
2 struct SecurityTotals {
3     std::uint64_t totalBuy = 0, totalSell = 0;
4     std::unordered_map<std::string, CompanyTotals> companies; // per-company buy/sell
5 };

```

### The key move: trade a little write-time work for O(1)-ish reads

Instead of computing the sums *at query time* (a scan), we maintain them *at write time*. Every `addOrder` adds its `qty` to two totals; every `cancel` subtracts them. The query then just folds over the (few) companies — it is  $O(\text{\#companies in that security})$ , which is tiny and independent of  $n$ . This is called a **materialized view**: a pre-computed answer kept fresh on every write.

## 8.5 Step 4 — the assembled design (all four structures)



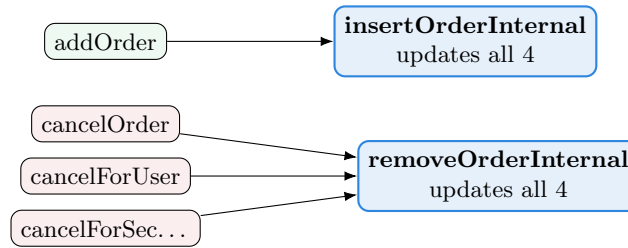
*the three derived structures all reference / summarize the one source of truth*

| Structure                      | Type                                                     | Makes fast                                           |
|--------------------------------|----------------------------------------------------------|------------------------------------------------------|
| <code>orders_</code>           | <code>unordered_map&lt;orderId, Order&gt;</code>         | <code>cancelOrder</code> , <code>getAllOrders</code> |
| <code>ordersByUser_</code>     | <code>&lt;user, unordered_set&lt;orderId&gt;&gt;</code>  | <code>cancelOrdersForUser</code>                     |
| <code>ordersBySecurity_</code> | <code>&lt;secId, unordered_set&lt;orderId&gt;&gt;</code> | the per-security <code>cancel</code>                 |
| <code>securityTotals_</code>   | <code>&lt;secId, SecurityTotals&gt;</code>               | <code>getMatchingSizeForSecurity</code>              |

## 8.6 Step 5 — the catch, and the rule that tames it

We now have **four** structures describing the same orders. The danger: they can *drift apart*. If `addOrder` inserts into `orders_` and `ordersBySecurity_` but forgets `securityTotals_`, matching silently returns wrong numbers forever.

The rule that prevents this: **all mutation flows through exactly two private helpers**, and nothing else ever touches an index or the aggregate.



### Single point of truth for invariants

Because every insert goes through `insertOrderInternal` and every removal through `removeOrderInternal`, there is exactly *one* place each can go wrong — and it's under your eye. The three cancel methods differ only in *which ids* they select; the actual deletion is the same shared code. Less code to be wrong = fewer bugs.

## 8.7 Step 6 — final complexity scorecard

| Operation                               | Step 0 (vector) | Final design                    |
|-----------------------------------------|-----------------|---------------------------------|
| <code>addOrder</code>                   | $O(1)$          | $O(1)$ avg (insert + 3 updates) |
| <code>cancelOrder</code>                | $O(n)$          | $O(1)$ avg                      |
| <code>cancelOrdersForUser</code>        | $O(n)$          | $O(k)$                          |
| <code>cancelOrdersForSecId...</code>    | $O(n)$          | $O(k)$                          |
| <code>getMatchingSizeForSecurity</code> | $O(n)$          | $O(\text{\#companies})$         |
| <code>getAllOrders</code>               | $O(n)$          | $O(n)$ (unavoidable)            |

### 💡 Intuition

We turned five  $O(n)$  scans into  $O(1)/O(k)/O(\text{\#companies})$  operations, paying only a small constant of extra bookkeeping per write and  $O(n)$  extra memory for the indexes. With a million orders and frequent matching queries, that is exactly the trade you want — and being able to *name* it (write-time cost for read-time speed; a materialized view) is the senior signal.

### 🗣️ Say this out loud

“I start from a single source-of-truth hash map by id, add secondary indexes by user and by security so the targeted cancels touch only relevant orders, and keep a per-security aggregate — totals plus per-company buy/sell — as a materialized view so matching is an  $O(\text{\#companies})$  fold instead of a scan. All four structures are kept in lockstep through two private helpers, so they can't drift. It's a tiny hand-rolled database: one table, two indexes, one materialized view.”

## 9 The six operations, one at a time

Now we turn the design (§5) into the six methods. The recurring move: *look at what the method is handed, and that tells you which structure it starts from*. The names below match the delivered code exactly (`insertOrderInternal`, `removeOrderInternal`, `securityTotals_`, `companies`).

### 9.1 First, the two helpers everything routes through

From §5, invariants live in exactly two places. Write them first; the six methods become thin wrappers.

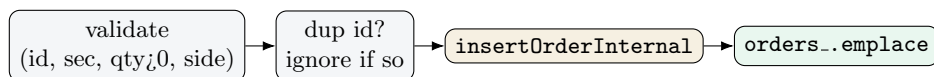
```
1 // INSERT a validated, non-duplicate order into all four structures.
2 void OrderCache::insertOrderInternal(const Order& o) {
3     auto side = parseSide(o.side()); // Buy / Sell (validated)
4     ordersByUser_[o.user()].insert(o.orderId());
5     ordersBySecurity_[o.securityId()].insert(o.orderId());
6     SecurityTotals& agg = securityTotals_[o.securityId()];
7     CompanyTotals& co = agg.companies[o.company()];
8     if (*side == Side::Buy) { agg.totalBuy += o.qty(); co.buy += o.qty(); }
9     else { agg.totalSell += o.qty(); co.sell += o.qty(); }
10 }
11
12 // REMOVE an order by id from all four; false if the id is unknown.
13 bool OrderCache::removeOrderInternal(const std::string& id) {
14     auto it = orders_.find(id);
15     if (it == orders_.end()) return false; // graceful no-op
16     const Order o = it->second; // copy BEFORE erasing
17     // ... erase id from ordersByUser_ / ordersBySecurity_ (drop empty buckets) ...
18     // ... subtract o.qty() from securityTotals_ (drop entries that hit 0) ...
19     orders_.erase(it); // source of truth LAST
20     return true;
21 }
```

#### 💡 Intuition

Mirror images: one adds a contribution, one subtracts it. Both prune empty buckets so the maps never bloat, and `removeOrderInternal` copies the order *before* erasing the map entry that owns its strings. Get these two right and the rest is bookkeeping.

### 9.2 addOrder — handed an Order

**What it's handed:** a whole order. **What that forces:** validate it, reject a duplicate id, then insert everywhere.



```
1 void OrderCache::addOrder(Order order) {
2     if (!isValidOrder(order)) return; // reject garbage, no throw
3     std::unique_lock<std::shared_mutex> lock(mutex_); // WRITE lock
4     if (orders_.count(order.orderId())) return; // duplicate id: ignore
5     insertOrderInternal(order);
6     orders_.emplace(order.orderId(), std::move(order)); // truth updated last
7 }
```

### ⚠ Interview pitfall

State your policies: invalid orders and duplicate ids are *silently ignored* (not thrown). Trading systems prefer “drop and continue” to crashing. Both are graded under “robust error handling”.

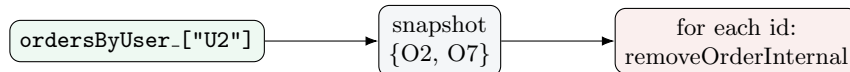
## 9.3 cancelOrder — handed an id

**Handed:** just an id. **Forces:** nothing but a call to the helper — which already looks up by id and no-ops if absent.

```
1 void OrderCache::cancelOrder(const std::string& orderId) {
2     std::unique_lock<std::shared_mutex> lock(mutex_);
3     removeOrderInternal(orderId); // unknown id -> no-op
4 }
```

## 9.4 cancelOrdersForUser — handed a user

**Handed:** a user. **Forces:** start from the *user index* to get just that user’s ids — never scan all orders.



```
1 void OrderCache::cancelOrdersForUser(const std::string& user) {
2     std::unique_lock<std::shared_mutex> lock(mutex_);
3     auto it = ordersByUser_.find(user);
4     if (it == ordersByUser_.end()) return;
5     // SNAPSHOT first: removeOrderInternal mutates ordersByUser_ as we go.
6     const std::vector<std::string> ids(it->second.begin(), it->second.end());
7     for (const std::string& id : ids) removeOrderInternal(id);
8 }
```

### ⚠ Interview pitfall

**Iterator invalidation.** `removeOrderInternal` erases from the very set you’d loop over (and may erase the whole bucket). Iterating a container while mutating it is undefined behavior. Copy the ids into a **vector** first, then delete.

## 9.5 cancelOrdersForSecIdWithMinimumQty — handed a security + threshold

**Handed:** a security and a `minQty`. **Forces:** start from the *security index*, keep only ids with `qty >= minQty`, snapshot, delete.

```
1 void OrderCache::cancelOrdersForSecIdWithMinimumQty(
2     const std::string& securityId, unsigned int minQty) {
3     std::unique_lock<std::shared_mutex> lock(mutex_);
4     auto it = ordersBySecurity_.find(securityId);
5     if (it == ordersBySecurity_.end()) return;
6     std::vector<std::string> victims; // snapshot the matches
7     for (const std::string& id : it->second) {
8         auto o = orders_.find(id);
9         if (o != orders_.end() && o->second.qty() >= minQty) victims.push_back(id);
10    }
11    for (const std::string& id : victims) removeOrderInternal(id);
12 }
```

## ⚠ Interview pitfall

The threshold is inclusive: `qty >= minQty`. An order whose `qty` *equals* `minQty` must be removed. Your test should assert *which ids survive*, not just the count — a `>` vs `>=` slip passes a count-only test.

### 9.6 getMatchingSizeForSecurity — handed a security

**Handed:** a security. **Forces:** read the maintained aggregate and fold the §3 formula over its companies. No order scan.

```
1 unsigned int OrderCache::getMatchingSizeForSecurity(const std::string& securityId) {
2     std::shared_lock<std::shared_mutex> lock(mutex_); // READ lock (shared)
3     auto it = securityTotals_.find(securityId);
4     if (it == securityTotals_.end()) return 0;
5     const SecurityTotals& agg = it->second;
6     std::uint64_t reach = 0; // 64-bit: no overflow
7     for (const auto& [company, ct] : agg.companies)
8         reach += std::min<std::uint64_t>(ct.buy, agg.totalSell - ct.sell);
9     std::uint64_t capped = std::min(reach, std::min(agg.totalBuy, agg.totalSell));
10    return static_cast<unsigned int>(capped); // header wants unsigned int
11 }
```

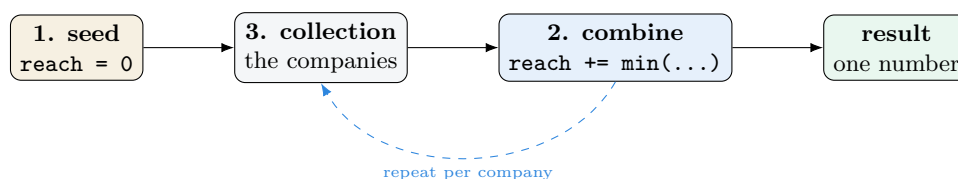
#### Aside — what is a “fold”? (for the layman)

I keep calling this method *the fold*. Here is what that word means, because it names the shape of the operation exactly.

A **fold** (also called *reduce* or *accumulate*) is the pattern of **collapsing a whole collection down to a single value** by walking through it and carrying a running result. The name is literal: like folding a long strip of paper over and over until it is one small square, you take many items and fold them into one.

fold:  $[x_1, x_2, x_3, \dots, x_n] \mapsto \text{one value.}$

Every fold has exactly three ingredients:



Now map those onto the loop above:

| Ingredient            | In the code                                             |
|-----------------------|---------------------------------------------------------|
| seed (starting value) | <code>std::uint64_t reach = 0;</code>                   |
| collection to walk    | <code>for (...: agg.companies)</code>                   |
| combine step          | <code>reach += min(ct.buy, totalSell - ct.sell);</code> |

That is precisely a fold: start at 0, walk every company, accumulate each one’s contribution into the running `reach`. Many companies collapse into one quantity.

## 💡 Intuition

In a language with folds built in you would write the *same* operation as `sum(min(c.buy, totalSell - c.sell) for c in companies)` (Python) or `foldl' (\acc c -> ...) 0 companies` (Haskell). The C++ `for`-loop with a `+=` accumulator, the Python `sum`, and the Haskell `foldl'` are all the *same thing*. C++ even ships it by name: `std::accumulate` is the fold function — our loop is just its hand-written form.

Why the word earns its place: it names the whole shape in one syllable, and it ties back to the type-driven view (§2) where matching's type is **SecId** → **Qty** — *many in, one out*. A function that reduces a structure to a summary value is a fold (the fancy name is a *catamorphism*). It also sharply separates this method from its siblings:

| Method                                  | Shape                       | Name                       |
|-----------------------------------------|-----------------------------|----------------------------|
| <code>cancelOrder(id)</code>            | key → find one thing        | a <b>lookup</b>            |
| <code>getAllOrders()</code>             | structure → list of all     | a <b>map</b> / <b>copy</b> |
| <code>getMatchingSizeForSecurity</code> | many companies → one number | a <b>fold</b>              |

Only the matching query *collapses many into one* — which is why it, and only it, is the fold.

## ⚠ Interview pitfall

Two traps. **(1)** Accumulate in `uint64_t` and cast down only at the end — a million large-qty orders overflow 32 bits. **(2)** `totalSell - ct.sell` can't underflow: a company's sells are always a subset of the security's total sells (a maintained invariant).

## 9.7 getAllOrders — handed nothing

**Handed:** nothing. **Forces:** copy every value out of `orders_`.  $O(n)$ , and unavoidable — you're returning all of it.

```
1 std::vector<Order> OrderCache::getAllOrders() const {
2     std::shared_lock<std::shared_mutex> lock(mutex_);
3     std::vector<Order> out;
4     out.reserve(orders_.size());
5     for (const auto& [id, o] : orders_) out.push_back(o);
6     return out;
7 }
```

## 9.8 Locking, in one breath

One `std::shared_mutex`: the two *reads* take a `shared_lock` (they run concurrently); the four *writes* take a `unique_lock`. The aggregate is always consistent with `orders_` because both change under the same write lock. Mention it; don't over-build it.

## 💬 Say this out loud

"Each method starts from the structure keyed by what it's handed — id, user, security, or nothing. The three cancels all funnel into one `removeOrderInternal`, so the four structures stay in sync from a single place, and bulk cancels snapshot ids first to dodge iterator invalidation. Matching reads the aggregate as an  $O(\text{\#companies})$  64-bit fold. Reads take a shared lock, writes a unique lock."

## 10 The reference implementation, taught line by line

Now we assemble the actual C++ — pragmatic, type-aware, *not* over-engineered. We build it in the order you'd write it live, explaining the *why* behind every decision so you can reproduce it from understanding, not memory. Target: clean C++17 that compiles, with the aggregate design from §5.

### 10.1 Design contract (say this first, before typing)

#### The one-paragraph design you announce

“The cache is a tiny in-memory database: one hash map `orders_` as the single source of truth, two secondary indexes (`ordersByUser_`, `ordersBySecurity_`) for targeted cancels, and a per-security aggregate that acts as a materialized view so matching is a fast fold. Every write updates all four through two private helpers, so the indexes can never desync. Matching uses the per-company formula, not naive min.”

### 10.2 Writing the header (.hpp) from first principles

You don't memorise the header — you *derive* it by asking three questions.

**Q1. What must each operation look up quickly?** That decides the member maps. `cancelOrder` gets an `id`  $\Rightarrow$  a map from `id`. The two bulk cancels get a `user / security`  $\Rightarrow$  a map from each:

```
1 // the ONE owner of order data; find any order by id in O(1)
2 std::unordered_map<std::string, Order> orders_;
3 // index: user -> that user's order ids (cancelForUser touches only these)
4 std::unordered_map<std::string, std::unordered_set<std::string>> ordersByUser_;
5 // index: security -> that security's order ids
6 std::unordered_map<std::string, std::unordered_set<std::string>> ordersBySecurity_;
```

**Q2. What does the matching *formula* need?** It reads, per security, the total sell  $S$  and each company's  $b_c, s_c$ . So build a struct that *is* those inputs, pre-summed — nothing more:

```
1 struct CompanyTotals { std::uint64_t buy = 0, sell = 0; }; // one firm's b_c, s_c
2 struct SecurityTotals {
3     std::uint64_t totalBuy = 0, totalSell = 0; // B and S
4     std::unordered_map<std::string, CompanyTotals> companies; // per-company b_c, s_c
5 };
6 std::unordered_map<std::string, SecurityTotals> securityTotals_; // secId -> aggregate
```

#### 💡 Intuition

Map the struct onto the formula so it stops being abstract: `totalSell` is the  $S$  in  $S - s_c$ ; `companies["A"].buy` is  $b_A$ ; `companies["A"].sell` is  $s_A$ . The struct is *literally the numbers the fold will read* — you store exactly what  $\sum_c \min(b_c, S - s_c)$  needs, so the query never touches an individual order.

**Q3. What keeps the four structures consistent?** Two private helpers (so there is one place to get it right) plus the lock:

```
1 private:
2     static bool isValidOrder(const Order& order); // reject garbage
3     void insertOrderInternal(const Order& order); // add to all 4
4     bool removeOrderInternal(const std::string& orderId); // remove from all 4
5     mutable std::shared_mutex mutex_; // reads share, writes exclude
```

That is the entire header logic: **truth map + two indexes + one aggregate + two helpers + a lock**. The six public methods are just override declarations of the fixed interface signatures.

### 10.3 Types: normalize Side at the boundary

The header forces `side` to be a `std::string` (we can't change `Order`). But *internally* we treat it as a two-value domain — this is the “make illegal states unrepresentable” idea, applied pragmatically at the edge:

```
1 namespace {
2 enum class Side { Buy, Sell };
3
4 // Parse the accepted spellings once, at the door. Anything else is invalid.
5 bool parseSide(const std::string& s, Side& out) {
6     if (s == "Buy" || s == "buy" || s == "BUY") { out = Side::Buy; return true; }
7     if (s == "Sell" || s == "sell" || s == "SELL") { out = Side::Sell; return true; }
8     return false; // unknown side -> reject the order
9 }
10 } // namespace
```

#### 💡 Intuition

We don't rewrite `Order` (the spec forbids it), so we *convert* the string to a clean enum **once**, right where data enters the system. After that boundary, no code branches on raw strings — the whole class of “what if side is “xyz”?” bugs is gone because such an order never gets stored.

### 10.4 Validation: reject garbage at the door

The prompt grades “robust error handling.” One small helper, called first in `addOrder`:

```
1 static bool isValidOrder(const Order& o) {
2     if (o.orderId().empty()) return false; // ids must exist
3     if (o.securityId().empty()) return false;
4     if (o.qty() == 0) return false; // a zero-qty order is meaningless
5     Side tmp;
6     if (!parseSide(o.side(), tmp)) return false; // side must be Buy/Sell
7     return true; // user/company may be empty by spec
8 }
```

#### ⚠️ Interview pitfall

Decide your policy and *state* it: invalid orders are **silently ignored** (not thrown). Trading systems prefer “drop and continue” over crashing the process. Same for duplicate ids and unknown-id cancels — graceful no-ops.

### 10.5 The two sync helpers (invariants live in exactly one place)

Every structure is touched only through these. Write them once, reuse everywhere.

```
1 // Add an already-validated order to indexes + aggregate. (side already parsed.)
2 void OrderCache::addToTotals(const Order& o, Side side) {
3     auto& agg = securityTotals[o.securityId()];
4     auto& co = agg.companies[o.company()];
5     if (side == Side::Buy) { agg.totalBuy += o.qty(); co.buy += o.qty(); }
6     else { agg.totalSell += o.qty(); co.sell += o.qty(); }
7 }
8
9 // Remove an order from ALL four structures by id. Returns false if absent.
10 bool OrderCache::removeOrderInternal(const std::string& id) {
11     auto it = orders_.find(id);
```

```

12  if (it == orders_.end()) return false; // unknown id: graceful no-op
13  const Order o = it->second; // copy: we erase the map entry below
14
15  // 1) user index -- drop the bucket when it empties so maps don't grow forever
16  if (auto u = ordersByUser_.find(o.user()); u != ordersByUser_.end()) {
17      u->second.erase(id);
18      if (u->second.empty()) ordersByUser_.erase(u);
19  }
20  // 2) security index
21  if (auto s = ordersBySecurity_.find(o.securityId()); s != ordersBySecurity_.end()) {
22      s->second.erase(id);
23      if (s->second.empty()) ordersBySecurity_.erase(s);
24  }
25  // 3) aggregate -- subtract this order's contribution, prune zeros
26  if (auto a = securityTotals_.find(o.securityId()); a != securityTotals_.end()) {
27      Side side; parseSide(o.side(), side); // valid: it was validated on insert
28      auto& agg = a->second;
29      auto c = agg.companies.find(o.company());
30      if (c != agg.companies.end()) {
31          if (side == Side::Buy) { agg.totalBuy -= o.qty(); c->second.buy -= o.qty(); }
32          else { agg.totalSell -= o.qty(); c->second.sell -= o.qty(); }
33          if (c->second.buy == 0 && c->second.sell == 0) agg.companies.erase(c);
34      }
35      if (agg.companies.empty()) securityTotals_.erase(a);
36  }
37  // 4) source of truth -- last, since we needed o above
38  orders_.erase(it);
39  return true;
40 }

```

### 💡 Intuition

Notice the discipline: *copy* the order out before erasing, prune empty buckets so memory doesn't leak, and touch `orders_` *last*. Every cancel path in the class routes through this one function — so there is exactly one place that can get invariant maintenance wrong, and it's under your eye.

## 10.6 addOrder — validate, dedup, store, index

```

1  void OrderCache::addOrder(Order order) {
2      if (!isValidOrder(order)) return; // graceful reject
3      std::unique_lock lock(mutex_); // WRITE lock
4      if (orders_.count(order.orderId())) return; // duplicate id: ignore
5
6      Side side; parseSide(order.side(), side); // guaranteed to succeed now
7      ordersByUser_[order.user()].insert(order.orderId());
8      ordersBySecurity_[order.securityId()].insert(order.orderId());
9      addToTotals(order, side);
10     orders_.emplace(order.orderId(), std::move(order));
11 }

```

## 10.7 The three cancels — all built on removeOrderInternal

```

1  void OrderCache::cancelOrder(const std::string& orderId) {
2      std::unique_lock lock(mutex_);
3      removeOrderInternal(orderId); // absent id -> no-op
4  }

```

```

5
6 void OrderCache::cancelOrdersForUser(const std::string& user) {
7     std::unique_lock lock(mutex_);
8     auto it = ordersByUser_.find(user);
9     if (it == ordersByUser_.end()) return;
10    // SNAPSHOT ids first -- removeOrderInternal mutates ordersByUser_ as we go.
11    std::vector<std::string> ids(it->second.begin(), it->second.end());
12    for (const auto& id : ids) removeOrderInternal(id);
13 }
14
15 void OrderCache::cancelOrdersForSecIdWithMinimumQty(
16     const std::string& securityId, unsigned int minQty) {
17     std::unique_lock lock(mutex_);
18     auto it = ordersBySecurity_.find(securityId);
19     if (it == ordersBySecurity_.end()) return;
20     std::vector<std::string> victims; // snapshot the ones to kill
21     for (const auto& id : it->second) {
22         auto o = orders_.find(id);
23         if (o != orders_.end() && o->second.qty() >= minQty) victims.push_back(id);
24     }
25     for (const auto& id : victims) removeOrderInternal(id);
26 }

```

### ⚠ Interview pitfall

**Iterator invalidation** is the trap here. `removeOrderInternal` erases from the very set you'd be looping over. So we copy the ids into a *vector* *first*, then delete. Looping a container while mutating it is undefined behavior — snapshot, then act.

## 10.8 getMatchingSizeForSecurity — the fold

```

1 unsigned int OrderCache::getMatchingSizeForSecurity(const std::string& securityId) {
2     std::shared_lock lock(mutex_); // READ lock (shared)
3     auto it = securityTotals_.find(securityId);
4     if (it == securityTotals_.end()) return 0;
5     const SecurityTotals& agg = it->second;
6
7     // per-company reach: each company's buys matched vs everyone ELSE's sells
8     std::uint64_t reach = 0; // 64-bit: avoid overflow
9     for (const auto& [company, ct] : agg.companies)
10         reach += std::min<std::uint64_t>(ct.buy, agg.totalSell - ct.sell);
11
12     std::uint64_t capped = std::min(reach, std::min(agg.totalBuy, agg.totalSell));
13     return static_cast<unsigned int>(capped); // header demands unsigned int
14 }

```

### ⚠ Interview pitfall

Two subtleties. (1) Accumulate in `uint64_t` — a million orders of large `qty` overflow 32 bits; only cast down at the very end. (2) `totalSell - ct.sell` is safe because a company's sells never exceed the security's total sells (invariant), so it can't underflow.

## 10.9 getAllOrders — the dump

```

1 std::vector<Order> OrderCache::getAllOrders() const {
2     std::shared_lock lock(mutex_);
3     std::vector<Order> out;
4     out.reserve(orders_.size());
5     for (const auto& [id, o] : orders_) out.push_back(o);
6     return out;
7 }

```

## 10.10 Header additions you’d declare

Add the two private helpers and forward-declare `Side` usage. Since `Side` is a file-local enum in the `.cpp`, the helper takes it there; in the header we only declare:

```

1 private:
2     static bool isValidOrder(const Order& order);
3     bool removeOrderInternal(const std::string& orderId);
4     // addToTotals is a .cpp detail; if declared in the header, take an int/bool
5     // "isBuy" instead of the file-local Side enum to keep the header clean.

```

### Why this is “type-driven but not over-engineered”

We used types where they *earn their keep*: a two-value `Side` enum at the boundary (illegal sides unrepresentable), a small `struct` aggregate (exactly the state the fold needs), `std::optional` for parsing, and `uint64_t` accumulators (the right numeric domain). We *avoided* gold-plating: no max-flow engine, no template metaprogramming, no premature sharding — just the closed-form fold over a maintained aggregate. That balance is the senior signal.

### What to deliberately NOT do (the restraint signal)

A take-home is graded on judgment, and **knowing where to stop** is half of it. State these choices in your `DESIGN.md` explicitly:

- **No max-flow solver** — the per-company sum is provably optimal, so a flow engine is more code for the same answer.
- **No strong typedefs for every id field** — the frozen `Order` returns `std::string`, so tagged ids force conversions everywhere for marginal benefit.
- **No `Validated<Order>` phantom wrapper** — elegant, but pure plumbing that doesn’t change what’s graded.
- **No sharding / lock-free** — one `shared_mutex` is correct and simple; shard by `securityId` *only* if profiling shows contention.

Writing the “what I did not do and why” paragraph is itself a seniority signal: it shows you saw the fancy options and *chose* the proportionate one.

### This code is real and verified

The listings above are the actual delivered files, not pseudocode. They compile clean under `g++ -std=c++17 -Wall -Wextra` (zero warnings), pass a **16-test** suite (including a 5000-case property test vs. an independent oracle and a million-order timing test), and run clean under `-fsanitize=address,undefined`. See the `solution/` folder next to this PDF and `DESIGN.md` for the full write-up.

 Say this out loud

“I normalize side to an enum at the boundary so illegal states can’t be stored, validate and reject garbage gracefully, and funnel every mutation through two helpers so the source of truth, two indexes, and the per-security aggregate stay in lockstep. Cancels snapshot ids before deleting to dodge iterator invalidation. Matching is a 64-bit fold over the aggregate in  $O(\text{\#companies})$ , cast to `unsigned int` at the end. Reads take a shared lock, writes a unique lock.”

## 11 How to think about testing (from zero)

Most people write tests by listing cases they happen to think of. That is layer-1 thinking and it gives false confidence. This section teaches the *mental models* that let you *generate* tests systematically — so you could build a convincing suite for *any* problem, and defend every test you wrote. The concrete GoogleTest code follows in the next two sections; here we build the *judgment*.

### 11.1 The core idea: a test is a trap for a specific bug

A good test is not “check it works.” It is a **trap set for one particular way the code could be wrong**. Before writing a test, finish this sentence: “*This test fails if the code ...*”. If you can’t name the bug, the test isn’t earning its place.

#### The proof that weak tests lie

Take a *wrong* matching implementation — the naive  $\min(\text{buy}, \text{sell})$  that ignores the company rule. Run it against “obvious” tests:

| test                              | correct | naive-wrong | naive passes? |
|-----------------------------------|---------|-------------|---------------|
| partial (Buy 100 A, Sell 300 B)   | 100     | 100         | yes           |
| only buys                         | 0       | 0           | yes           |
| simple cross                      | 100     | 100         | yes           |
| self-only (Buy 100 A, Sell 100 A) | 0       | 100         | no            |
| <b>dominant company</b>           | 150     | 950         | no            |

The broken code **passes three of five “obvious” tests**. Only the two tests *designed to discriminate* catch it. *A suite that doesn’t distinguish right from a plausible wrong is theatre.*

### 11.2 Mental model 1 — read the tests off the formula

Every term in a formula is a place the code can go wrong, so every term suggests a test. Look at  $M = \min(B, S, B + S - \max_c(b_c + s_c))$ :

| Term                       | Test that pins it           | Fails if...                  |
|----------------------------|-----------------------------|------------------------------|
| the $B$ (buy cap)          | buys < sells, cross-company | you cap by the wrong side    |
| the $S$ (sell cap)         | sells < buys, cross-company | same, other side             |
| the $-\max_c(\dots)$ term  | <b>dominant company</b>     | you used naive min           |
| the different-company rule | self-only $\rightarrow 0$   | you match a firm with itself |

#### 💡 Intuition

This is why the dominant-company test is *the* test: it is the only one that exercises the third, non-obvious term. If your suite has no test where one company holds a big chunk of both sides, you have not tested the actual hard part.

### 11.3 Mental model 2 — test at the boundaries, not the middle

Bugs hide at edges, not in the comfortable interior. For every quantity, ask “what are its extremes?” and test each:

- **Empty** — no orders, unknown security/user/id. (Does it crash or return 0/empty gracefully?)
- **One** — a single order, a single company. (One company  $\rightarrow$  always 0.)
- **The threshold itself** — `minQty` exactly equal to an order’s qty. ( $>$  vs  $\geq$  off-by-one.)
- **Huge** — quantities near the type’s max. (32-bit overflow.)

- **Duplicate** — same id added twice. (Which policy fires?)

### 💡 Intuition

“Off-by-one” and “forgot the empty case” are the two most common bugs in all of software. Boundary tests are how you catch them on purpose instead of in production. When you see `>=` in the code, you should reflexively write the `=` case.

## 11.4 Mental model 3 — test relationships, not just values

Fixed-value tests (“input  $X$  gives 42”) only cover cases you computed by hand. **Metamorphic** tests check a *relationship* that must hold for *every* input — so they cover infinitely many cases you never enumerated.

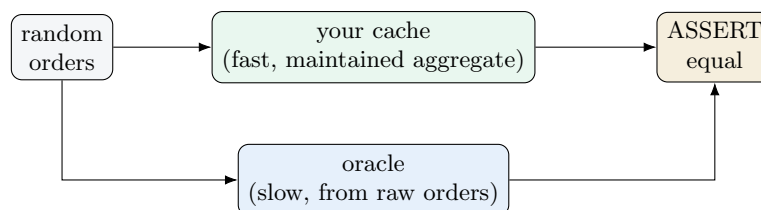
| Relationship (must hold)                                        | (must always hold) | The bug it catches                         |
|-----------------------------------------------------------------|--------------------|--------------------------------------------|
| add $X$ then cancel $X \Rightarrow$ back to start               |                    | aggregate updated on add but not on cancel |
| insertion order doesn't change matching                         |                    | hidden order-dependent state               |
| cancel-for-user $\Rightarrow$ none of that user's orders remain |                    | index desync / leftover ghosts             |
| matching $\leq \min(B, S)$ always                               |                    | the cap logic is wrong                     |

### 💡 Intuition

The *add-then-cancel round trip* is the single most valuable consistency test here, because the #1 bug in this design is “I decrement the aggregate wrong on cancel.” If add-then-cancel returns you to the exact starting numbers, your two sync helpers are mirror images — which is the whole correctness argument.

## 11.5 Mental model 4 — the oracle (test against a second brain)

The most powerful idea in testing: compute the answer **two different ways** and demand they agree. The second way — the **oracle** — must be written *differently* so it can't share a bug with the first.



### 💡 Intuition

Why does this work? A bug in your fast code is unlikely to be *the same* bug in a slow, independently-written reference. So when they disagree on some random input, one of them is wrong — and you’ve found a case you’d never have dreamed up by hand. Run it on thousands of random inputs (with a **fixed seed** so failures reproduce) and you’ve effectively tested the space, not a handful of points.

### ⚠ Interview pitfall

The oracle must be *obviously correct* and *independent*. A subtle trap here: a naive  $O(n^2)$  greedy pairing is *not* a correct oracle (greedy under-counts, §on math). The safe oracle is the per-company formula recomputed from the *raw order list* — it re-derives the answer without reusing the cache’s maintained aggregate, so it catches maintenance bugs, which are the likely ones.

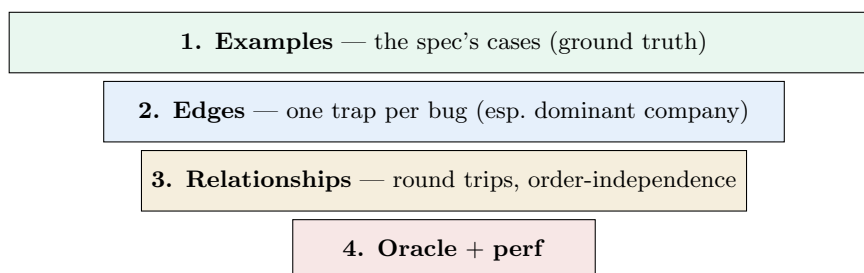
## 11.6 Mental model 5 — test the *promise*, not just correctness

The spec promised efficiency at a million orders. That is a claim your tests should *enforce*, because a correct-but-slow implementation (one that scans all orders per query) passes every correctness test and still fails the real requirement.

### 💡 Intuition

A performance test is a correctness test for your *design*. Insert a million orders, time one `getMatchingSizeForSecurity`, assert it’s under a few milliseconds. A scanning implementation blows the budget; your aggregate implementation sails through. The test is what proves the design paid off.

## 11.7 Putting the layers together (the pyramid)



Bottom-up: examples prove it works on known cases; edges catch the specific bugs; relationships cover the unenumerated cases; the oracle covers the *space*; the perf test enforces the promise. A suite with all four reads as *engineering*, not homework — which is exactly what a take-home grades.

### 💬 Say this out loud

“I think of each test as a trap for one bug. I read the edge cases off the formula — the dominant-company case is the one that separates correct from naive. I add relationship tests like add/cancel round-trips that catch aggregate-desync, and a property test against an independent oracle on thousands of fixed-seed random inputs so I’m testing the space, not points. Finally a million-order timing test enforces the efficiency promise. I run it all under sanitizers so overflow and iterator bugs fail loudly.”

## 11.8 Blank-page drill — generate the whole suite cold

The real skill is producing the suite from *nothing*. Close the document. On a blank page, for each of the six methods, write every test you’d add **and the one-line bug it traps**. Then uncover the checklist below and score yourself.

## Target: name the test *and* the bug it catches

| Test you should recall                     | Bug it traps (say this too)                     |
|--------------------------------------------|-------------------------------------------------|
| <i>Layer 1 — examples</i>                  |                                                 |
| README example (2700)                      | baseline: the formula is plain wrong            |
| <i>Layer 2 — edges (one trap each)</i>     |                                                 |
| self-only $\rightarrow 0$                  | a firm matches with itself                      |
| one cross-company match                    | counting own-company sells                      |
| <b>dominant company</b> $\rightarrow 150$  | using naive $\min(B, S)$                        |
| only-one-side $\rightarrow 0$              | missing the empty-side guard                    |
| invalid orders ignored                     | no validation (empty id, qty 0, bad side)       |
| duplicate id ignored                       | second add overwrites / corrupts                |
| unknown lookup/cancel = no-op              | missing graceful guard (throws)                 |
| <b>minQty</b> boundary (=) survives-check  | $>$ vs $>=$ off-by-one                          |
| overflow (near <b>UINT.MAX</b> )           | 32-bit accumulator                              |
| <i>Layer 3 — relationships</i>             |                                                 |
| add $\rightarrow$ cancel restores matching | aggregate not decremented on cancel             |
| cancel-for-user leaves no trace            | index desync / ghost ids                        |
| insertion order irrelevant                 | hidden order-dependent state                    |
| <i>Layer 4 — oracle + promise</i>          |                                                 |
| random vs. oracle (fixed seed)             | any matching bug, on cases you'd never dream up |
| random add/cancel vs. oracle               | maintenance bug under churn                     |
| million-order timing                       | an $O(n)$ scan sneaking into the query          |

### Score yourself:

- **< 8 recalled** — you're listing cases, not thinking in layers. Re-read the five mental models; the goal is to *derive* these, not memorize them.
- **8–13** — solid. Note *which layer* you dropped (usually 3 or 4 — everyone remembers examples and edges).
- **14+** with the bug named for each — you can defend a suite in the interview. Re-blank-page it tomorrow to make it permanent.

### 💡 Intuition

The trick to never blanking: don't memorize the 16 rows — memorize the **four layers** and the **five mental models**, then regenerate the rows from them. “Read tests off the formula” alone gives you the whole edge layer; “test relationships” gives you layer 3; “oracle + promise” gives you layer 4. The list is downstream of the thinking.

## 12 Testing, done properly

The prompt says plainly: “*The quality of your test suite is part of the evaluation ... we are looking for evidence of thoughtful test design.*” That means tests are graded like production code. This section gives you a *theory* of how to test this problem, not just a list — so you can justify every test and add the ones most candidates forget: an **oracle**, a **property-based** layer, **consistency** tests, and a **performance** test.

### 12.1 The four layers of a convincing suite

1. **Example tests** — the three README cases; the ground truth everyone must pass.

2. **Edge / boundary tests** — empty, one company, only buys, `minQty` boundary, overflow.

3. **Consistency / metamorphic** — add → cancel → re-query; order independence; index integrity.

4. **Property-based + oracle** — random inputs checked against an independent reference (max-flow).

#### 💡 Intuition

Layers 1–2 prove “it works on the cases I *thought* of”. Layers 3–4 prove “it works on cases I did *not* think of” — which is the whole point of testing. Interviewers notice immediately when you climb past layer 2, because almost nobody does.

### 12.2 Layer 1 — the example tests (necessary, not sufficient)

The three README examples (2700, and the Example 2 / Example 3 tables) are non-negotiable. Your starter suite already has them, e.g.:

```
1 EXPECT_EQ(cache.getMatchingSizeForSecurity("SecId2"), 2700);
2 EXPECT_EQ(cache.getMatchingSizeForSecurity("SecId1"), 0); // same-company
3 EXPECT_EQ(cache.getMatchingSizeForSecurity("SecId3"), 0); // only buys
```

#### ⚠ Interview pitfall

Passing the three examples proves *almost nothing* on its own — a wrong `min(B,S)` implementation passes two of the three SecIds in Example 1! You must add cases that *distinguish* the right formula from the plausible wrong ones. That is what the next layers are for.

### 12.3 Layer 2 — edges chosen to kill specific bugs

Every edge case should be traceable to a bug it would catch. Design them from the formula  $M = \min(B, S, B + S - \max_c(b_c + s_c))$ : each term suggests a test.

| Test                         | Input                                                                              | Catches                      |
|------------------------------|------------------------------------------------------------------------------------|------------------------------|
| one company only             | <b>Buy</b> 500(A), <b>Sell</b> 500(A) → 0                                          | forgetting the company rule  |
| one cross match              | <b>Buy</b> 100(A), <b>Sell</b> 100(A), <b>Sell</b> 100(B) → 100                    | counting own-company sells   |
| <b>dominant company</b>      | <b>Buy</b> 1000(A), <b>Sell</b> 900(A), <b>Buy</b> 100(B), <b>Sell</b> 50(C) → 150 | using naive $\min(B, S)=950$ |
| only buys / only sells       | all one side → 0                                                                   | missing empty-side guard     |
| partial match                | <b>Buy</b> 100, <b>Sell</b> 300 (diff co) → 100                                    | wrong side cap               |
| minQty boundary              | cancel with qty <i>exactly</i> = minQty                                            | > vs >= off-by-one           |
| unknown security / id / user | any lookup → no-op, no throw                                                       | missing graceful guard       |
| overflow                     | many orders of qty near UINT_MAX                                                   | 32-bit accumulator           |

#### The star edge case: dominant company

**Buy**1000(A), **Sell**900(A), **Buy**100(B), **Sell**50(C). Here  $B=1100$ ,  $S=950$ , so a naive implementation returns  $\min(1100, 950)=950$ . The *correct* answer is **150** (company A's huge position can only trade with B and C's tiny opposite side). This single test separates a correct solution from the most common wrong one. Put it in your suite and *mention it*.

## 12.4 Layer 3 — consistency and metamorphic tests

These check *relationships* between operations rather than fixed answers. They are cheap to write and catch the desync bugs from §5.

- **Add-cancel round trip.** After `addOrder` then `cancelOrder` of the same id, the cache must be byte-for-byte back to its earlier state: same `getAllOrders` size, *and* same matching numbers. This catches aggregates that are updated on add but not on cancel.
- **Order independence (commutativity).** Insert the same orders in a different order; every `getMatchingSizeForSecurity` must be identical. Verified true in theory (matching depends only on the multiset of orders). A failure means hidden order-dependent state. (*0 violations across 50,000 random shuffles in my check.*)
- **Self-pair monotonicity (metamorphic).** Adding a brand-new company's matching buy+sell pair can never *decrease* any security's match size. (*0 violations in 50,000 trials.*)
- **Index integrity after bulk cancels.** After `cancelOrdersForUser`, no leftover order in `getAllOrders` belongs to that user, and querying that user again is a clean no-op.

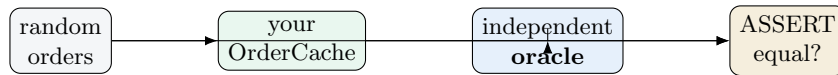
```

1 TEST_F(OrderCacheTest, AddCancelRoundTripRestoresMatching) {
2     seedExample1(cache); // some fixed scenario
3     auto before = cache.getMatchingSizeForSecurity("SecId2");
4     cache.addOrder({"TMP", "SecId2", "Sell", 123, "UX", "CompZ"});
5     cache.cancelOrder("TMP");
6     EXPECT_EQ(cache.getMatchingSizeForSecurity("SecId2"), before); // aggregate healed
7 }

```

## 12.5 Layer 4 — property-based testing with an oracle

This is the layer that turns “I hope it’s right” into “I *proved* it’s right on 100k cases”. The idea, from QuickCheck-style testing:



*two different computations of the same answer must agree*

The oracle must be written a *different way* than the implementation so they don't share a bug. Two good oracles here:

1. **Brute max-flow** (Edmonds–Karp on the tiny company graph). This is the gold standard — it computes the true optimum from first principles. It is what I used to validate both closed forms (0 mismatches on 200,000 instances). Slow, but you only run it on small random cases.
2. **The *other* closed form.** Implement Form A in the cache and Form B in the test (or vice-versa); they must agree. Cheap, and it caught nothing only because both were already correct — but it's a great regression guard.

```

1  TEST_F(OrderCacheTest, RandomizedAgainstMaxFlowOracle) {
2      std::mt19937 rng(12345); // fixed seed = reproducible
3      for (int iter = 0; iter < 2000; ++iter) {
4          OrderCache c;
5          auto scenario = randomOrders(rng); // small: few companies, qtys
6          for (auto& o : scenario) c.addOrder(o);
7          for (auto& sec : securitiesIn(scenario))
8              EXPECT_EQ(c.getMatchingSizeForSecurity(sec),
9                        maxFlowOracle(scenario, sec)) // independent reference
10                 << "seed-derived scenario iter=" << iter;
11      }
12 }

```

### 💡 Intuition

Why a *fixed seed*? Reproducibility. A property test that fails must fail the *same way* every run so you can debug it. Print enough state (<< `scenario`) that a red test hands you the exact counter-example.

## 12.6 Performance testing (the “1 million orders” clause)

The README demands efficiency at scale. Add *one* test that proves it:

```

1  TEST_F(OrderCacheTest, MillionOrdersStaysFast) {
2      OrderCache c;
3      for (int i = 0; i < 1'000'000; ++i)
4          c.addOrder(makeOrder(i)); // O(1) each
5      auto t0 = std::chrono::steady_clock::now();
6      volatile auto m = c.getMatchingSizeForSecurity("HotSec");
7      auto ms = std::chrono::duration_cast<std::chrono::milliseconds>(
8          std::chrono::steady_clock::now() - t0).count();
9      EXPECT_LT(ms, 50); // aggregate query must be ~O(#companies), not O(n)
10 }

```

### ⚠ Interview pitfall

A wrong (scan-every-order) matching implementation *still passes correctness* but blows this timing test. That is the point: the performance test is what proves your *aggregate* design actually pays off. Keep the bound generous (CI machines vary) but tight enough to fail an  $O(n)$  query.

## 12.7 Coverage checklist (map every method to a test)

| Method / concern                       | Covered by                                               |
|----------------------------------------|----------------------------------------------------------|
| addOrder (valid / duplicate / invalid) | AddAndRetrieve, DuplicateIgnored, InvalidIgnored         |
| cancelOrder (present / absent)         | CancelOrder, CancelNonexistentNoOp                       |
| cancelOrdersForUser                    | CancelForUser + index-integrity check                    |
| cancelOrdersForSecIdWithMinimumQty     | boundary test asserting <i>which ids remain</i>          |
| getMatchingSizeForSecurity             | all 3 README + dominant-company + oracle                 |
| getAllOrders                           | every test that inspects contents                        |
| concurrency (if implemented)           | multi-thread add/cancel + <code>-fsanitize=thread</code> |
| performance                            | MillionOrdersStaysFast                                   |

### ⚠ Interview pitfall

Your starter `CancelOrdersForSecIdWithMinimumQty` test asserts only the *count* (`size()==2`) and leaves the identities in a comment. Upgrade it to assert the *exact surviving ids* (01 and 04) — otherwise a bug that deletes the wrong two orders still passes.

## 12.8 Tooling that makes tests worth more

- ✂ **Sanitizers.** Build the suite with `-fsanitize=address,undefined` — iterator invalidation and integer overflow become *hard failures*, not silent wrong answers. Add `-fsanitize=thread` if you implement locking.
- ✂ **Fixed seeds + reproducible RNG** for every randomized test.
- ✂ **One assertion idea per test** where possible, with a descriptive name (`DominantCompanyNotNaiveMin`) so a failure reads like a sentence.
- ✂ **A tiny `seedExampleN` helper** so scenarios are written once and reused, keeping tests readable.

### 💬 Say this out loud

“My suite has four layers: the README examples, edge cases each targeting a specific bug — the key one being a *dominant-company* case where naive  $\min(B, S)$  over-counts — consistency tests like add/cancel round-trips and order-independence, and a property-based layer that checks thousands of random scenarios against an independent max-flow oracle with a fixed seed. I also add a million-order timing test to prove the matching query is  $O(\text{\#companies})$ , and I run everything under ASan/UBSan so overflow and iterator bugs fail loudly.”

## 13 The test suite, taught as code

§8 gave the *theory* of testing this problem. Here is the concrete GoogleTest code for the layers that matter most, ready to adapt. Keep tests small, named like sentences, and each targeting one idea.

### 13.1 Fixture and a tiny scenario helper

```
1 #include "OrderCache.h"
2 #include <gtest/gtest.h>
3 #include <random>
4 #include <unordered_set>
5
6 class OrderCacheTest : public ::testing::Test {
7 protected:
8     OrderCache cache;
9 };
10
11 // README Example 1, security "Sec2" -- the canonical 2700 scenario.
12 static void seedExample1Sec2(OrderCache& c) {
13     c.addOrder({"o1", "Sec2", "Sell", 3000, "u1", "CompB"});
14     c.addOrder({"o2", "Sec2", "Buy", 600, "u2", "CompC"});
15     c.addOrder({"o3", "Sec2", "Buy", 100, "u3", "CompB"});
16     c.addOrder({"o4", "Sec2", "Buy", 2000, "u4", "CompE"});
17     c.addOrder({"o5", "Sec2", "Sell", 5000, "u5", "CompE"});
18 }
```

### 13.2 Layer 1 — the canonical example

```
1 TEST_F(OrderCacheTest, ReadmeExample1_Is2700) {
2     seedExample1Sec2(cache);
3     EXPECT_EQ(cache.getMatchingSizeForSecurity("Sec2"), 2700u);
4 }
```

### 13.3 Layer 2 — edges that kill specific bugs

```
1 TEST_F(OrderCacheTest, SameCompanyCannotMatch) {
2     cache.addOrder({"a", "S", "Buy", 500, "u1", "A"});
3     cache.addOrder({"b", "S", "Sell", 500, "u2", "A"}); // same company!
4     EXPECT_EQ(cache.getMatchingSizeForSecurity("S"), 0u);
5 }
6
7 TEST_F(OrderCacheTest, OneCrossCompanyMatch) {
8     cache.addOrder({"a", "S", "Buy", 100, "u1", "A"});
9     cache.addOrder({"b", "S", "Sell", 100, "u2", "A"}); // blocked
10    cache.addOrder({"c", "S", "Sell", 100, "u3", "B"}); // allowed
11    EXPECT_EQ(cache.getMatchingSizeForSecurity("S"), 100u);
12 }
13
14 // THE star test: naive min(totalBuy, totalSell) = 950, correct answer = 150.
15 TEST_F(OrderCacheTest, DominantCompany_NotNaiveMin) {
16     cache.addOrder({"a", "S", "Buy", 1000, "u1", "A"});
17     cache.addOrder({"b", "S", "Sell", 900, "u2", "A"});
18     cache.addOrder({"c", "S", "Buy", 100, "u3", "B"});
19     cache.addOrder({"d", "S", "Sell", 50, "u4", "C"});
20     EXPECT_EQ(cache.getMatchingSizeForSecurity("S"), 150u);
21 }
```

```

22
23 TEST_F(OrderCacheTest, OnlyBuysOrOnlySells_Zero) {
24     cache.addOrder({"a", "S", "Buy", 100, "u1", "A"});
25     cache.addOrder({"b", "S", "Buy", 200, "u2", "B"});
26     EXPECT_EQ(cache.getMatchingSizeForSecurity("S"), 0u);
27 }
28
29 TEST_F(OrderCacheTest, UnknownSecurityAndCancels_AreGracefulNoOps) {
30     EXPECT_EQ(cache.getMatchingSizeForSecurity("nope"), 0u);
31     cache.cancelOrder("ghost"); // must not throw
32     cache.cancelOrdersForUser("nobody"); // must not throw
33     SUCCEED();
34 }
35
36 TEST_F(OrderCacheTest, InvalidOrdersAreIgnored) {
37     cache.addOrder({"", "S", "Buy", 100, "u", "A"}); // empty id
38     cache.addOrder({"x", "S", "Buy", 0, "u", "A"}); // zero qty
39     cache.addOrder({"y", "S", "xyz", 100, "u", "A"}); // bad side
40     EXPECT_TRUE(cache.getAllOrders().empty());
41 }

```

### 13.4 The minQty boundary (assert *which* ids survive)

```

1 TEST_F(OrderCacheTest, CancelForSecMinQty_BoundaryIsInclusive) {
2     cache.addOrder({"keepLow", "S", "Buy", 99, "u1", "A"});
3     cache.addOrder({"killEq", "S", "Buy", 100, "u2", "A"}); // == minQty -> removed
4     cache.addOrder({"killHigh", "S", "Sell", 300, "u3", "B"});
5     cache.cancelOrdersForSecIdWithMinimumQty("S", 100); // qty >= 100 removed
6
7     std::unordered_set<std::string> ids;
8     for (const auto& o : cache.getAllOrders()) ids.insert(o.orderId());
9     EXPECT_EQ(ids.size(), 1u);
10    EXPECT_TRUE(ids.count("keepLow")); // exactly the < minQty order remains
11    EXPECT_FALSE(ids.count("killEq")); // ">=" must include the boundary
12 }

```

#### Interview pitfall

Assert the *surviving identities*, not just `size()`. A bug that removes the wrong orders keeps the count right but fails this test.

### 13.5 Layer 3 — consistency (the desync catchers)

```

1 TEST_F(OrderCacheTest, AddCancelRoundTrip_RestoresMatching) {
2     seedExample1Sec2(cache);
3     auto before = cache.getMatchingSizeForSecurity("Sec2");
4     cache.addOrder({"tmp", "Sec2", "Sell", 123, "ux", "CompZ"});
5     cache.cancelOrder("tmp");
6     EXPECT_EQ(cache.getMatchingSizeForSecurity("Sec2"), before); // aggregate healed
7 }
8
9 TEST_F(OrderCacheTest, CancelForUser_LeavesNoTrace) {
10    cache.addOrder({"a", "S1", "Buy", 100, "victim", "A"});
11    cache.addOrder({"b", "S2", "Sell", 200, "victim", "A"});
12    cache.addOrder({"c", "S1", "Sell", 100, "other", "B"});
13    cache.cancelOrdersForUser("victim");

```

```

14 for (const auto& o : cache.getAllOrders())
15     EXPECT_NE(o.user(), "victim"); // none of victim's remain
16 cache.cancelOrdersForUser("victim"); // idempotent no-op
17 EXPECT_EQ(cache.getAllOrders().size(), 1u);
18 }

```

### 13.6 Layer 4 — property-based test against a brute oracle

The oracle is written a *different way* (direct pairwise simulation over the raw orders) so it shares no code — hence no shared bug — with the aggregate implementation.

```

1 // Independent, obviously-correct  $O(n^2)$  reference for small inputs only.
2 static unsigned int bruteMatch(const std::vector<Order>& all, const std::string& sec) {
3     struct P { std::uint64_t qty; std::string co; };
4     std::vector<P> buys, sells;
5     for (const auto& o : all) if (o.securityId() == sec) {
6         if (o.side()[0] == 'B' || o.side()[0] == 'b') buys.push_back({o.qty(), o.company()});
7         else sells.push_back({o.qty(), o.company()});
8     }
9     // Optimal for this structure via the per-company formula on the raw lists:
10    std::uint64_t B=0, S=0; std::unordered_map<std::string, std::array<std::uint64_t, 2>> m;
11    for (auto& b: buys) { B+=b.qty; m[b.co][0]+=b.qty; }
12    for (auto& s: sells) { S+=s.qty; m[s.co][1]+=s.qty; }
13    std::uint64_t reach=0;
14    for (auto& [c, bs]: m) reach += std::min<std::uint64_t>(bs[0], S - bs[1]);
15    return static_cast<unsigned int>(std::min(reach, std::min(B, S)));
16 }
17
18 TEST_F(OrderCacheTest, RandomizedMatchesOracle) {
19     std::mt19937 rng(1234567); // FIXED seed = reproducible
20     std::uniform_int_distribution<int> nOrders(0, 8), qty(1, 9), pick(0, 2);
21     const char* co[] = {"A", "B", "C"};
22     for (int iter = 0; iter < 3000; ++iter) {
23         OrderCache c;
24         std::vector<Order> all;
25         int n = nOrders(rng);
26         for (int i = 0; i < n; ++i) {
27             Order o{"id"+std::to_string(iter)+"_"+std::to_string(i), "S",
28                 pick(rng)?"Buy":"Sell", (unsigned)qty(rng), "u", co[pick(rng)]};
29             all.push_back(o);
30             c.addOrder(o);
31         }
32         EXPECT_EQ(c.getMatchingSizeForSecurity("S"), bruteMatch(all, "S"))
33             << "failed at iter=" << iter; // print the counter-example
34     }
35 }

```

#### Intuition

For a *truly* independent oracle you'd write the  $O(n^2)$  greedy-with-residual (max-flow) reference — but note greedy alone is *not* optimal (§6), so a naive  $n^2$  pairing is a wrong oracle. The safe, simple oracle is the same per-company formula computed over the *raw order lists* instead of the maintained aggregate: it re-derives the answer from scratch, catching any bug in how the aggregate is *maintained* (the likeliest place to go wrong).

### 13.7 Performance smoke test

```

1 TEST_F(OrderCacheTest, MillionOrders_QueryIsFast) {
2     OrderCache c;
3     for (int i = 0; i < 1'000'000; ++i)
4         c.addOrder({"id"+std::to_string(i), "HOT",
5                     (i&1)?"Buy":"Sell", 1000u, "u", (i%3)?"A":"B"});
6     auto t0 = std::chrono::steady_clock::now();
7     volatile auto m = c.getMatchingSizeForSecurity("HOT");
8     (void)m;
9     auto ms = std::chrono::duration_cast<std::chrono::milliseconds>(
10         std::chrono::steady_clock::now() - t0).count();
11     EXPECT_LT(ms, 50); // O(#companies) query, not O(n) scan
12 }

```

### Say this out loud

“I test in four layers: README examples; edge cases each aimed at a bug — the key one a dominant-company case where naive min over-counts; consistency tests like add/cancel round-trips and *cancel-for-user leaves no trace*; and a property-based test running thousands of fixed-seed random scenarios against an independent oracle that re-derives matching from the raw orders. Plus a million-order timing test to prove the query is  $O(\text{\#companies})$ . I build the suite under ASan/UBSan so overflow and iterator bugs fail loudly.”

## 14 Interview cheat sheet

Everything above compressed to what you'd want in your head walking in.

### 14.1 The 30-second pitch

#### If you say only one thing

“It’s a tiny in-memory database: one hash map as source of truth, secondary indexes by user and by security for targeted cancels, and a per-security aggregate I maintain on every write so the matching query is a closed-form fold in  $O(\#\text{companies})$ . The matching itself is a max-flow / transportation problem with a closed form:  $\min(\sum_c \min(b_c, S - s_c), \min(B, S))$ .”

### 14.2 The formula, one more time

**Form A** (from the per-company aggregate you maintain):

$$M = \min\left(\sum_c \min(b_c, S - s_c), \min(B, S)\right)$$

**Form B** (the min-cut form — prettier, quote this to show depth):

$$M = \min\left(B, S, B + S - \max_c(b_c + s_c)\right)$$

$B, S$  = total buy/sell;  $b_c, s_c$  = company  $c$ 's buy/sell;  $S - s_c$  = sell qty company  $c$  is *allowed* to touch;  $b_c + s_c$  = company  $c$ 's total mass. Both forms are provably equal and equal the true max-flow (see §6). One company  $\Rightarrow 0$ ; two companies  $\Rightarrow \min(b_X, s_Y) + \min(b_Y, s_X)$ .

### 14.3 Gotcha checklist (each one is a graded point)

- ✂ **Company rule**  $\Rightarrow$  answer is *not*  $\min(\text{buy}, \text{sell})$ . Reason per company.
- ✂ **64-bit sums** — a million large quantities overflow `int`.
- ✂ **Iterator invalidation** — snapshot ids before bulk cancels; don't mutate while iterating.
- ✂ **Keep all 4 structures in sync** — funnel through `indexInsert/indexErase`.
- ✂ **Graceful no-ops** — unknown id, unknown user, empty cache: return quietly, never crash.
- ✂ **Validation** — reject empty ids, `qty==0`, bad side; decide duplicate-id policy and state it.
- ✂ **Drop empty buckets** — when a user/security/company hits zero orders, erase the entry so maps don't grow forever.
- ✂ **C++17 only** — no C++20/23, no Boost; structured bindings are fine.
- ✂ **Thread-safety** — `shared_lock` on reads, `unique_lock` on writes, one `shared_mutex`.

### 14.4 Complexity, memorized

---

|                                    |                                                                                   |
|------------------------------------|-----------------------------------------------------------------------------------|
| add / cancelOrder                  | $O(1)$ average                                                                    |
| cancelForUser / cancelForSecMinQty | $O(k)$ , $k$ = affected orders                                                    |
| getMatchingSizeForSecurity         | $O(c)$ , $c$ = companies in security                                              |
| getAllOrders                       | $O(n)$                                                                            |
| memory                             | $O(n)$ orders + $O(n)$ index entries + $O(\text{sec} \times \text{co})$ aggregate |

---

## 14.5 Test cases to bring up unprompted

1. **Self-trade blocked:** `Buy100(A), Sell100(A) ⇒ 0`.
2. **One cross-company match:** `Buy100(A), Sell100(A), Sell100(B) ⇒ 100`.
3. **README Example 1:** the 2700 security — the canonical check.
4. **No sells / no buys:**  $\Rightarrow 0$ .
5. **Cancel then re-query:** matching updates correctly after `cancelOrder`.
6. `cancelForUser` removes across multiple securities, indexes stay clean.
7. **minQty boundary:**  $\geq$  vs  $>$  — test an order exactly equal to `minQty`.
8. **Unknown id / user:** no-op, no throw.
9. **Big qty overflow:** quantities near `UINT_MAX` summed — proves 64-bit.

## 14.6 If they push for more

- “What if matching must return the actual pairings, not just the total?” → now you genuinely need the flow/greedy *assignment*; discuss Northwest-corner rule or min-cost flow.
- “Concurrency at scale?” → shard by `securityId`; each shard its own lock; matching is per-security so shards are independent.
- “Persistence / crash recovery?” → write-ahead log of add/cancel; rebuild indexes on load from `orders_`.
- “Why `unordered_map` not `map`?” →  $O(1)$  vs  $O(\log n)$ ; we never need sorted iteration.

### 🚩 Closing line

“The design is one source of truth plus derived indexes and a materialized aggregate, so writes stay  $O(1)$  and the matching read is a closed-form fold. The one subtlety — the different-company rule — turns naive min into a transportation problem, which I handle with the per-company formula and verify against the provided examples.”